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Mission: Teach 1 million - Mathematics for AI / Quantitative Finance

444+ Problems in Probability

for AI and Quantitative Finance

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1. Classical Probability and Combinatorics

1.1. Level 1

1.1.1 Problem 1

You are playing a 2D game where you are trapped within a 6×6 grid and navigates via points on the grid. You starts at the bottom left: point $(0,0)$. You can only move up and right. How many possible paths can you take to get to the top right point: $(6,6)$?

1.1.2 Problem 2

8 standard 52-card decks are combined together and shuffled well. Find the probability that 3 cards drawn uniformly at random from this deck form a three card straight. Note that we do not count a straight flush in this.

1.1.3 Problem 3

Calculate the probability that when we roll 4 fair 6-sided dice, the sum of their upfaces is 20.

1.1.4 Problem 4

How many times does the digit 9 occur when counting 1 to 1000? Note that the number 919, for example, would count as 2 9s.

1.1.5 Problem 5

Let $x_0 = 0$ and let $x_1, \dots, x_{10}$ satisfy that $|x_i - x_{i-1}| = 1$ for $1 \leq i \leq 10$ and $x_{10} = 4$. How many such sequences are there satisfying these conditions?

1.1.6 Problem 6

A computer is interpreting an 8-bit binary string, which consists of 8 characters that are either 1 or 0. How many such strings begin with 1 or end with the two characters 00?

1.1.7 Problem 7

Erica intends to construct a subset T of $S = \{I, J, K, L, M, N\}$, but if she is unsure about including an element x of S in T , she will write x in bold and include it in T . For example, $\{I, J\}$, $\{J, K, L\}$, and $\{I, J, M, N\}$ are valid examples of T , while $\{I, J, J, K\}$ is not. Find the total number of such sets T that Erica can construct.

1.1.8 Problem 8

A fair 6-sided die, a standard roulette wheel (has 38 values with two zeroes and 1–36), and a standard deck of cards are in a casino. Letting Aces count as ones, what is the probability that when the roulette wheel is spun, the die is rolled, and a card is dealt from the deck, all of them show the same value?

1.1.9 Problem 9

7 people sit around a circular table uniformly at random. All of them have a distinct age. Find the probability that they sit down at the table in age order. Note that the ages can be increasing in either the clockwise or counter-clockwise directions.

1.1.10 Problem 10

There are 14 boys standing around in a circle, of which one of them is Charlie. 10 girls walk up and join the circle at random to form a larger circle. Each girl must be between 2 boys. If the girls arrange such that all ways of forming a larger circle are equally likely, find the probability Charlie is still standing between two boys.

1.1.11 Problem 11

Evaluate

$$12\binom{3}{3} + 11\binom{4}{3} + 10\binom{5}{3} + \cdots + 2\binom{13}{3} + \binom{14}{3}.$$

1.1.12 Problem 12

You have 4 fair coins in front of you. You flip all 4 coins and observe their outcomes. You can then choose to turn over any 2 coins at a time, but you cannot turn just 1 coin without turning another. (A turn moves a coin from heads to tails or tails to heads). You can repeat turning over pairs of coins as many times as you want. The goal of the game is to get the most amount of heads, assuming you play optimally, what is the expected number of heads you will have?

1.1.13 Problem 13

Consider coloring the 6 sides of a cube each a distinct color among red, blue, green, yellow, purple, and pink. Find the probability that the sides that are colored red, green, and blue all share a vertex of the cube (i.e. corner) in common.

1.1.14 Problem 14

How many distinct ways can you arrange 3 red balls, 7 blue balls, and 9 green balls in a line?

1.1.15 Problem 15

10 pairs of socks, each with a distinct color, are in a drawer. You draw out 2 socks at random. Find the probability that you obtain a matching pair.

1.1.16 Problem 16

What is the total number of paths from $(0, 0, 0)$ to $(3, 4, 5)$ in 3D space, given that we can move one unit to the right, forward, or up at each step?

1.1.17 Problem 17

A room of four couples greet each other by shaking hands. If each person shakes the hand of every other person besides their partner, how many handshakes occur?

1.1.18 Problem 18

How many $3 \times 3 \times 3$ cubes can be fit inside a cube of dimensions $20 \times 20 \times 20$?

1.1.19 Problem 19

Four numbers from $\{1, 2, \dots, 9\}$ are selected with repetition allowed. Find the probability that the product of the four integers has a units digit of 1, 3, 7, or 9.

1.1.20 Problem 20

What is the probability of taking out 13 cards from a well shuffled standard 52 card deck and getting exactly 2 Aces? Round your answer to the nearest hundredths.

1.1.21 Problem 21

Let F_n be the n th Fibonacci number. Compute $\frac{F_1 + F_2 + F_3 + \dots + F_{300}}{F_3 + F_6 + F_9 + \dots + F_{300}}$.

1.1.22 Problem 22

You roll a fair 6-sided die twice. Calculate the probability that the value of the first roll is strictly less than the value of the second roll.

1.1.23 Problem 23

A child has a pack of 6 green and 10 yellow jellybeans. The child wants to eat 4 jellybeans, so they grab 4 jellybeans one-by-one uniformly at random without replacement from the pack. Find the probability that the first two and last two jellybeans agree in colors (not necessarily in order). For example, *GYYG* and *GGGG* are both valid, but *GGYG* is not.

1.1.24 Problem 24

How many distinguishable permutations of *LOLLIPOP* are there that start and end with the same letter? (e.g *LOLIPOPL* is valid as it starts and ends with L)

1.1.25 Problem 25

An archer is shooting at targets. There are 4 columns of identical targets, each with 2, 3, 3, and 4 targets stacked vertically, respectively. The archer first selects a column and then shoots the lowest-hanging target in that column that is not broken. Once a column is out of targets, it is no longer able to be selected. In how many ways can the archer break all the targets?

1.1.26 Problem 26

How many subsets of $\{1, 2, \dots, 10\}$ contain exactly 2 elements from $\{1, 2, 3\}$?

1.1.27 Problem 27

How many subsets of $\{1, 2, \dots, 10\}$ contain exactly 2 elements from $\{1, 2, 3\}$ AND only have odd elements?

1.1.28 Problem 28

Marissa has a deck of 40 cards: 4 1's, 4 2's, $\dots$, 4 10's. A matching pair—2 cards with the same number—is removed from the deck without replacement. If Marissa randomly draws 2 more cards, what is the probability that she draws another matching pair?

1.1.29 Problem 29

You have a box of 6 marbles of distinct colors, two of which are blue and green. You draw 4 marbles one-at-a-time with replacement. Find the probability that there is at least one green and one blue marble among the 4 you selected.

1.1.30 Problem 30

Seven people with distinct ages randomly sit down at a circular table with seven seats. What is the probability that the people sit themselves in increasing order of age, irrespective of direction?

1.1.31 Problem 31

You place three dots along four edges of a square at random. What is the probability that the dots lie on distinct edges?

1.1.32 Problem 32

How many ways can you sum three positive integers to get a total of 25?

1.1.33 Problem 33

A standard fair 6-sided die is rolled 6 times. We'll define a successful roll as rolling a prime number; otherwise, it's a failure. What is the probability that there are exactly 3 successful rolls?

1.1.34 Problem 34

You roll two fair 6-sided dice. Each of the two dice have the first 6 prime numbers on the sides. Find the probability that the sum of the two upfaces is also prime?

1.1.35 Problem 35

Two distinct prime integers between 1 and 20, inclusive, are selected uniformly at random. Find the probability their sum is even.

1.1.36 Problem 36

We have 6 weights of mass $201, 202, \dots, 206$ grams. We randomly arrange 3 weights on each side of a scale. Find the probability that the side with the 206 gram weight is heavier.

1.1.37 Problem 37

Your building a tower that's 12 blocks tall comprised of 1-tall or 2-tall blocks. Assuming you have at least 12 of each type of block, how many different tower configurations can be created?

1.1.38 Problem 38

Mo Salah is lining up to take a penalty kick. He decides to perform $3n$ motions. Each motion is either 2 steps forward or 1 step backwards. Find the probability that after 9 motions, Salah is exactly where he started.

1.1.39 Problem 39

There are 3 shiny and 4 dull pennies in a box. Pennies are drawn from the box without replacement. Compute the probability that it takes more than 4 draws for the third shiny penny to appear.

1.1.40 Problem 40

A deck of cards is shuffled well. The cards are dealt one-by-one, until the two of hearts appears. Find the probability that no kings, queens or jacks appear before the two of hearts.

1.1.41 Problem 41

The integers $1 - 100$ are split into two groups containing 30 and 70 numbers respectively. Compute the probability that numbers 6 and 66 are in different groups.

1.1.42 Problem 42

You have 5 students that you want to see 3 times each over the next month. There are 15 time slots available for the students to select from. However, one of the students can't attend in the first time slot. How many different ways can these students be scheduled?

1.1.43 Problem 43

There are 8 points in some space, no three of which lie on the same line. Matt and Aaron each uniformly at random select 3 points in the space. Matt and Aaron respectively draw a triangle whose vertices are the 3 points they selected. What is the probability they end up with the same triangle?

1.1.44 Problem 44

There are 6 points in space, no three of which are collinear (meaning none of them lie on the same line). Alice and Bob each randomly select 3 distinct points to form triangles. Although Alice and Bob individually must pick 3 different points to make a triangle, they both can choose the same points in their group of 3. What is the probability that their triangles share at least one vertex?

1.1.45 Problem 45

Jimmy has 8 friends. He will only invite 5 to his party. However, two of his friends don't get along and they refuse to attend together. How many possible combinations of guests are possible given this constraint?

1.1.46 Problem 46

How many valid arithmetic expressions using only $*$ and $+$ are there that include the values 123456789 in order? Some examples include $1 + 2 + 3456 + 7 * 89$, $123 + 456 * 78 + 9$, and 123456789 .

1.1.47 Problem 47

How many positive integers at most 10000 have the number 0 somewhere?

1.2. Level 2

1.2.1 Problem 1

You have 5 pairs of gloves that each have a distinct number 1–5. The 10 gloves are randomly paired up. Find the probability that the gloves are paired up such that the values of any pair differ by at most 1.

1.2.2 Problem 2

Consider the set of 10 consecutive integers $\{1, 2, \dots, 10\}$. How many subsets contain exactly 1 pair of consecutive integers? For example, $\{3, 5, 6, 9\}$ contains exactly 1 pair of consecutive integers.

1.2.3 Problem 3

You're at a big auction for a treasure chest with 10 participants. The chest's value is determined by flipping 20 fair coins behind a curtain. For each head, \$1 is added to the chest. Participants cannot see the flips but know how the value is set. Each participant submits a non-negative integer bid. The chest is awarded to a random person among the highest bidders, who pays their bid. Everyone else gets their bid returned. The goal is to count the number of pure Nash equilibria in this game, assuming utility for money is linear between $\{0\}$ and $\{10\}$.

1.2.4 Note:

In a pure Nash equilibrium, each player must choose a single bid with probability 1 (i.e., no mixed strategies).

1.2.5 Problem 4

Two turtles, Tort and Bort, are going to perform independent simple symmetric random walks on the integers starting at positions 0 and 4, respectively. Compute the probability after 10 steps, Tort and Bort are back at their initial positions and that Tort was strictly behind Bort at all 10 steps. The answer will be in the form

$$\frac{\binom{a}{b}^p - \binom{c}{d}^p}{2^r}$$

where $b \leq \frac{a}{2}$ and $d \leq \frac{c}{2}$. Find $a^2 + b^2 + c^2 + d^2 + p^2 + r^2$.

1.2.6 Problem 5

We flip a fair coin until we obtain our first heads. If the first heads occurs on the k th flip, we are given k balls. We put them into 3 bins labeled 1, 2, and 3 at random. Find the probability that none of the three bins are empty.

1.2.7 Problem 6

38 people are waiting to see a tennis match. The entrance fee is \$5 to enter. 19 of the people only have \$10 bills, while the other 19 only have \$5 bills. The cashier currently has no change. If the 38 people arrange themselves in a line completely at random, what is the probability that all of the people can successfully purchase a ticket without reordering in line?

1.2.8 Problem 7

Five intervals are each selected according to the following procedure: two points are sampled from $U[0, 1]$, the larger becoming the right endpoint and the smaller becoming the left endpoint. What is the probability that there exists a point of intersection between all five intervals? The answer can be written as a simplified fraction of the form $\frac{p}{q}$. Find $p + q$.

1.2.9 Problem 8

A frog is hopping between 8 lily pads labeled 1 – 8 from left to right. The frog starts on lily pad 1. At each turn, the frog hops to a lily pad that it has not been to yet. Find the probability the frog made exactly 1 leftwards hop after visiting all of the stones.

1.2.10 Problem 9

Consider an array of white unit squares arranged in a rectangular grid with r rows of unit squares and c columns of unit squares. What is the smallest possible value of c such that, if we shade s unit squares in each column black, then there must necessarily be some row with at least t black unit squares? You should receive a function $f(r, t, s)$. Evaluate $f(30, 21, 14)$.

1.2.11 Problem 10

Let $S = \{1, 2, 3, \dots, 22\}$. Find the number of ways in which S can be partitioned into eleven subsets such that each subset contains exactly two elements of S and the absolute difference between the two elements of a subset is 1 or 11.

1.2.12 Problem 11

Consider a 2023×2023 grid. The grid squares from the middle row are shaded in (this is the 1012th row). What is the probability that a randomly selected rectangle contains at least one shaded square?

1.2.13 Problem 12

Find the probability of obtaining a sum of 10 when rolling three fair 6-sided dice.

1.2.14 Problem 13

You are playing a 2D game where your character is trapped in a 6×6 grid. Your character starts at $(0, 0)$ and can only move up and right. There are two power-ups located at $(2, 3)$ and $(4, 6)$. How many possible paths can your character take to get to $(6, 6)$ such that it can collect at least one power-up?

1.2.15 Problem 14

Find the sum of all 9-digit integers using all of the digits $1 - 9$. The answer is in the form $k \cdot a! \cdot (10^b - 1)$ for a, b , and k integers such that a and b are maximal. Find abk .

1.2.16 Problem 15

Assume the dealer holds a fair deck of 52 cards. She shuffles the deck and gives each of you and your three friends at the table 13 cards. What is the probability that each of you receive an ace?

1.2.17 Problem 16

$\frac{5}{12}$ of concert attendees are adults. A bus carrying 50 people arrives at the concert. Now, $\frac{11}{25}$ of concert attendees are adults. What is the minimum number of adults who could have been at the concert before the bus arrived?

1.2.18 Problem 17

How many 6-sided dice with values on each side in the set $\{1, 2, 3, 4, 5, 6\}$ are there with the property that when rolled twice, for each integer $2 \leq k \leq 12$, there is positive probability that the sum is exactly k ? Note that not every value in the set necessarily needs to be used and that two dice are considered indistinguishable if they contain the exactly same amount of faces corresponding to each value in the set, regardless of the labelling of the sides. Assume each side appears with equal probability.

1.2.19 Problem 18

How many ways can 7 white balls and 8 black balls be distributed into 5 bowls such that each bowl receives exactly 3 balls?

1.2.20 Problem 19

Amy has a bowl of 5 red cherries and 8 purple cherries. She takes out cherries one at a time until there are no red cherries left. What is the probability that the bowl is empty?

1.2.21 Problem 20

There are 15 boys and 10 girls in a class. They line up in a row in random order. What is the expected amount of times a boy and a girl are standing next to each other? Ex: *BGBBGGGBGGBBBBBG* has 7 occurrences where a boy and girl are next to each other.

1.2.22 Problem 21

Max is building a tower from three distinct colors of building blocks (red, blue, and yellow). He has 7 blocks of each color and randomly picks a block out at a time to build his tower of 5 blocks in height. What is the probability the tower has composed of exactly two different colors?

1.2.23 Problem 22

A chess tournament has 2^n players, each with a distinct rating. Assume that the player with the higher rating always wins against a lower rated opponent with probability $0.5 < p < 1$. The winner proceeds to the subsequent round. Since the tournament's structure resembles that of a knockout bracket, n total rounds are played, including the final. The probability that the second-highest rated player defeats the highest rated player in the final round can be expressed as a function of n and p . Determine this function evaluated at $n = 4, p = \frac{3}{4}$.

1.2.24 Problem 23

Four fair coins appear in front of you. You flip all four at once and observe the outcomes of the coins. After seeing the outcomes, you may flip any pair of tails again. You may not flip a single coin without flipping another. You can iterate this process as many times as there are at least two tails to flip. Find the expected number of heads that we end up with after this process is complete.

1.2.25 Problem 24

Four fair coins appear in front of you. You flip all four at once and observe the outcomes of the coins. After seeing the outcomes, you may flip any pair of tails again. You may not flip a single coin without flipping another. You can iterate this process as many times as there are at least two tails to flip. Find the expected number of coin flips needed until you are unable to better your position.

1.2.26 Problem 25

You flip 4 fair coins and observe the outcomes. After seeing the outcomes, you select any pair of coins to reconsider. From the pair, one coin can be turned over while the other must be flipped again. If you already have 3 heads, you still must perform both actions. You iterate this process until you end up with all 4 coins being heads. A movement of a coin is when you either turn over or flip it. Find the expected number of coin movements needed to obtain 4 heads Note: the initial flip of the 4 coins do count to coin movements.

1.2.27 Problem 26

A coin is flipped 10 times and the outcomes are recorded. Find the probability that any tails occur only in consecutive pairs. For example, with 4 flips, $TTHH$, $TTTT$, and $HHHH$ are both valid, but $HTHH$ and $TTTH$ are not valid.

1.2.28 Problem 27

You have a 5×5 checkerboard. You randomly place a checker in the center 3×3 grid. The checkers can move in two equally random ways: up to the left, and up to the right. If the checker is on the boundary, then it can only make valid moves. What is the probability that the checker ends up in the top-left or top-right corner?

1.2.29 Problem 28

How many distinct ways can you label a 6 sided dice if you wipe off all the numbers? Arrangements that can be formed by rotating the die around are not considered distinct.

1.2.30 Problem 29

You roll two fair dice. What is the probability that the highest value rolled is a four?

1.2.31 Problem 30

8 fair standard 6-sided dice are thrown, the probability that the sum of the numbers on the top faces is 12 can be written as $\frac{x}{6^8}$. What is x ?

1.2.32 Problem 31

A particle starts at $(4, 4)$. Each turn, it moves either 1 unit in the $-x$ direction, 1 unit in the $-y$ direction, or 1 unit in each of the $-x$ and $-y$ directions. During a turn, the particle decides how it will move randomly such that the probability of each option is $\frac{1}{3}$. The particle repeatedly takes turns until it hits the x or y axes for the first time. Compute the probability that the particle hits the origin.

1.2.33 Problem 32

Suppose you flip a fair coin 10 times. We say a certain outcome (heads or tails) is leading after n flips if there are strictly larger than $\frac{n}{2}$ flips of that outcome within the first n flips. Find the probability that the outcome of the first flip of the coin is leading after 10 flips.

1.2.34 Problem 33

10 kids are really hungry! Their babysitter has 12 units of food to give. However, she decides she only wants to give 4 of the children food. How many ways can she distribute the food units such that 6 of the children are hungry (receive no food), and the other 4 children receive at least 1 unit of food each?

1.2.35 Problem 34

Suppose that you are trying to calculate the terms in the Fibonacci sequence. You calculate the values of $F_0, F_1, \dots, F_{99}$ correctly. However, you say that the value of F_{100} is $G_{100} = 1 + F_{100}$. Assuming you carry over this error into your subsequent calculations of terms of $F_{101}, \dots, F_{110}$, let G_{110} be the value that you calculate for F_{110} . Find $G_{110} - F_{110}$.

1.2.36 Problem 35

7 people uniformly at random select a random integer from the set $\{1, 2, \dots, 7\}$. Given that nobody selected 7, find the probability that 7 is the only value nobody selected.

1.2.37 Problem 36

How many terms are there in the expansion of $(x_1 + x_2 + x_3 + x_4 + x_5 + x_6)^{18}$ after all like terms have been combined?

1.2.38 Problem 37

Calculate the sum of all multinomial coefficients $\binom{7}{b_1, b_2, b_3, b_4}$, where $b_1 + \dots + b_4 = 7$ and every $b_i \geq 0$ is an integer.

1.2.39 Problem 38

How many permutations of $\{1, 2, \dots, 7\}$ have no adjacent even digits?

1.2.40 Problem 39

How many 10 digit numbers are there whose digits are prime and whose product of its digits is exactly 100000?

1.2.41 Problem 40

Find the sum of all binomial coefficients $\binom{n}{k}$ with n and k being positive odd integers at most 1000. Your answer is in the form $\frac{a^b - c}{d}$, where a, b, c , and d are integers and b is maximal. Find $abcd$.

1.2.42 Problem 41

How many non-negative odd integer solutions are there to the equation $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 96$? The answer is in the form $\binom{n}{k}$ for $k < \frac{n}{2}$. Find nk .

1.2.43 Problem 42

Suppose we have the values $1 - 6$ in a bowl. We draw them without replacement, noting the order in which we selected them. We multiply the first two values together, the next two values together, and the last two values together. Lastly, we add the three products above. How many unique sums are possible?

1.2.44 Problem 43

Let $f : \mathbb{R}^{10} \rightarrow \mathbb{R}$, where $f = f(x_1, \dots, x_{10})$ is a smooth function (i.e. partial derivatives exist of all orders). How many 5th order partial derivatives does f have? For example, $\frac{\partial f}{\partial x_1^5}$ and $\frac{\partial f}{\partial x_1^2 \partial x_2^2 \partial x_6}$ are both 5th order partial derivatives.

1.2.45 Problem 44

The n th row of Pascal's Triangle contains three successive entries with ratio $3 : 4 : 5$. What is the smallest value of n ?

1.2.46 Problem 45

Ten students of distinct heights are lining up for a picture. The photographer requires that the two tallest students stand in the two center positions and that the remaining students line up such that the heights strictly descend outwards. How many ways can the students line up?

1.2.47 Problem 46

50 people are competing in a ping pong tournament where there is only one ping pong table. The competitors are numbered 1 through 50. Suppose that if two competitors meet, the one with the larger number wins. Two competitors are chosen at random, and the loser is removed from the tournament. The winner moves on to the next round, where their opponent is chosen at random. This process is repeated until one person is left (a

total 49 rounds will be played). Compute the probability that competitor 49 is still in the tournament after the first 10 rounds have been played.

1.2.48 Problem 47

Pizza Hut gives new users a 7-digit code when they sign up. There are no restrictions on how many times a digit can be used, and digits are allowed to start with 0. How many such codes are non-decreasing? In other words, each digit in the passcode is at least as large as the last. 1122345 and 9999999 are such examples, whereas 2345657 is not.

1.2.49 Problem 48

Determine the maximum number of disjoint regions into which the plane can be divided by 10 non-parallel lines. For instance, a single line divides the plane into 2 regions.

1.2.50 Problem 49

A poker hand consists of five cards from a fair deck of 52 cards. What is the probability that you have a single pair (two cards of the same value and the other three cards of different unique values)?

1.2.51 Problem 50

A poker hand consists of five cards from a fair deck of 52 cards. What is the probability that you have three-of-a-kind (three cards of the same rank and two cards of two other ranks)?

1.2.52 Problem 51

m is randomly selected from $\{111, 133, 155, 177, 199\}$. n is randomly selected from $\{2004, 2005, \dots\}$. What is the probability that m^n has a 1 as the units digit?

1.2.53 Problem 52

There is a 3×3 grid of light bulbs, where each light bulbs are turned on with probability $\frac{1}{2}$. Find the probability that no two adjacent (grid cells that share a common side) light bulbs are powered on.

1.2.54 Problem 53

A quant firm wants to protect their IP. The firm only has 7 employees. They want to ensure that a safe can be opened only when at least 4 of them want to open it. Therefore, they put some locks on the safe. All of the locks must be unlocked to open the safe. Let n be the minimum number of locks needed to achieve this goal. Let m be the number of keys each of the 7 people carries. Find $10m + n$.

1.2.55 Problem 54

10 contestants are arranged into a line on a bridge and in front of them lay ten left tiles and ten right tiles side by side. In order to cross the bridge, the contestants must cross 10 tiles, and at each step, the person in front must pick either the left or right tile to step on. However, for each left & right tile pair, there is exactly one sturdy tile and one faulty tile, but the contestants cannot tell them apart. The contestants cross the bridge in their assigned order with the first person picking either the left or right tile, and continuing to lead unless either a faulty tile is picked (resulting in elimination) or person one reaches the other side. If the first person is eliminated before reaching the other side, the person second in line assumes the lead picking until he/she is eliminated (or reaches the other side), and so on. The winner of the game is the *first* person to reach the other side. Let p_i be the probability that the i th contestant in the line wins. Find $\max_i p_i$.

1.2.56 Problem 55

In a standard 3×3 tic tac toe grid, a player wins when they have 3 of their symbol (either X or O) in a row, column, or upon one of the two diagonals, yielding 8 possible winning paths. How many winning paths are there in $4 \times 4 \times 4$ 3D tic tac toe grid? The winning condition is that a player must have 4 of their symbol in a line, which includes diagonally and vertically down.

2. Axiomatic and Conditional Probability

2.1. Level 1

2.1.1 Problem 1

3 fair coins are flipped, and you are told that at least one of the coins showed heads. Find the probability exactly one coin showed heads.

2.1.2 Problem 2

Jed is rolling two fair 6-sided dice repeatedly. Find the probability that Jed obtains a sum of 4 before a sum of 2?

2.1.3 Problem 3

Suppose we draw 3 numbers from a $\text{Unif}[0, 1]$ distribution independently. We then sort the numbers in ascending order. What is the probability that the median is less than $2/3$?

2.1.4 Problem 4

Annie and Bob are playing a coin flipping game where they each have biased coins. Annie's coin lands on heads with probability $\frac{1}{3}$. Bob's coin lands on heads with probability $\frac{2}{5}$. Annie and Bob take turns tossing coins; Annie goes first. Given the first person to get a heads wins what is the probability that Annie wins?

2.1.5 Problem 5

There are three ants each on their own side of an equilateral triangle. Each picks one adjacent vertex to move to with equal probability. What is the probability that no two ants will intersect at a corner?

2.1.6 Problem 6

A hospital nursery has 3 baby boys and an unknown number of baby girls. A mother just gave birth to a child of unknown gender and it is added to the nursery. The doctor then picks up a random child in the nursery and it is a baby boy. Assuming that births of boys and girls are equally likely, find the probability that the mother just gave birth to a boy.

2.1.7 Problem 7

There are four red cherries and five purple cherries in bowl A . There are six red cherries and three purple cherries in bowl B . Jenny randomly transfers a cherry from bowl A into bowl B . Then, she randomly picks a cherry from bowl B to eat. What is the probability that she eats a red cherry?

2.1.8 Problem 8

Yugi is tossing darts at a dartboard. Each throw has a 10% probability of hitting the bullseye, with each throw being independent of the others. How many throws, denoted as n , does Yugi need to make in order to ensure that he has at least a 90% probability of hitting the bullseye at least once during those n throws?

2.1.9 Problem 9

On a given busy intersection, the probability of at least one car crash in a 1 hour time interval is $\frac{8}{9}$. Assuming that car crashes occur independently of one another and at a constant rate throughout time, find the probability of at least one car crash in a 30 minute interval.

2.1.10 Problem 10

As a good test taker, on a multiple choice exam with 5 options per question, Gabe either knows the answer to a question beforehand or chooses an answer completely at random. If he knows the answer beforehand, he selects the correct answer. If the probability that Gabe knows the answer to any given question is 0.6, find the probability that an answer that was correct was one for which he knew the answer.

2.1.11 Problem 11

We suppose that the number of children that a family has in the USA is $N \sim \text{Poisson}(\lambda)$ for some λ . We prescribe a prior distribution of $\lambda \sim \text{Gamma}(32, 10)$ to the distribution, where we use the Shape-Rate parameterization. We then observe 34 families who have a total of 100 kids combined. Find the posterior mean for λ .

2.1.12 Problem 12

Vishnu and Jason have 1000 and 500 in their respective bank accounts. They flip a fair coin repeatedly. If heads, Vishnu gives Jason 1. Else, Jason gives Vishnu 1. What is the probability that Jason runs out of money first?

2.1.13 Problem 13

A chess board consists of 2 inch by 2 inch squares. We toss a coin (diameter of 1 inch) that lands somewhere on the board randomly. What is the probability that the coin is completely within one of the 2 inch by 2 inch squares (not on more than 1 square)?

2.1.14 Problem 14

Alice and Bob are playing a game where they toss a fair coin. The first person to flip both a heads and a tails (not necessarily in that order) wins the game. Alice starts and then gives the coin to Bob and the cycle repeats until one of them has flipped both a heads and a tails. What is the probability that Alice wins this game?

2.1.15 Problem 15

Two players are playing a game by rolling a single standard die. Player 1 rolls first. If the top surface of the die shows 1 or 2, then he wins. If not, then Player 2 will roll the die. If the top surface of the die shows 3, 4, 5, or 6, then he wins. Else, the game continues until there is a winner. Let p_1 and p_2 represent the probabilities that the two players win, respectively. Find $\max\{p_1, p_2\}$.

2.1.16 Problem 16

A point is uniformly at random selected from a cone with radius 1 and height 2. Find the probability that the point is at most a height 1 away from the circular base.

2.1.17 Problem 17

A bag has 19 fair coins and one double-headed coin. We pick a coin uniformly at random from the bag. What is the minimum number of times this coin would need to appear heads in a row to be at least 95% certain that the coin we have is the double-headed coin?

2.1.18 Problem 18

You roll a die 5 times and record the outcomes. What is the probability of rolling at least 3 consecutive 1s in those 5 rolls?

2.1.19 Problem 19

A fair die is tossed 16 times, and the values are recorded in the order they appear. The values are then treated as individual digits and appended together. What is the probability that the resulting number is divisible by 8?

2.1.20 Problem 20

Your on a random walk along a discrete line consisting of 11 points, ranging from 0 to 10. You have an equal probability of moving either up or down with each step. Once you reach either 10 or 0, you must halt your movement. If you begin at point 6, what is the probability of reaching 10 before reaching 0?

2.1.21 Problem 21

Suppose we have a bag of distinct biased coins such that a randomly chosen coin's probability of heads has density $f(t) = 2t$ on $(0, 1)$. You choose a coin at random, toss it 6 times, and win if you get 6 heads. What is your probability of winning?

2.1.22 Problem 22

A fair 6-sided die has the values 1 – 6 on the sides. The value of one of the sides (selected uniformly at random) is decreased by 1. Find the probability an even value is rolled on this die.

2.1.23 Problem 23

Three bowls are presented to you. One has two blue balls, one has two red balls, and the last has one red ball and one blue ball. You select one bowl uniformly at random and note that you drew a blue ball from it. You then draw from the same bowl again after replacing the ball you selected. Find the probability that you would draw a blue ball on this draw.

2.1.24 Problem 24

Abby and Ben are playing a coin flipping game. Abby has a coin with probability p of showing heads on each flip and Ben has a fair coin. Abby flips her coin first, and then they alternate until one of them shows a heads. The person who flips the first heads is declared the winner. If it is known that Abby and Ben have equal chances of winning the game, find p .

2.1.25 Problem 25

Dave and Buster each have a fair 6-sided die, and they roll simultaneously. Buster wins if they roll the same value. If Buster rolls a strictly larger value than Dave, then Dave wins. However, if Dave rolls a strictly larger value than Buster, they must re-roll until one of the other two outcomes occurs. What is the probability that Buster wins this game?

2.1.26 Problem 26

Ahmed flips a fair coin until he obtains two consecutive heads or tails for the first time. What is the probability that Ahmed flips the coin an even amount of times.

2.1.27 Problem 27

Emilia has $n + 1$ fair coins and Ron has n fair coins. What is the probability that Emilia will flip more heads than Ron if both flip all of their coins?

2.1.28 Problem 28

Emma has two 6-sided dice with the values $1 - 6$ on the sides. One of the dice is fair, while the other is loaded such that each side appears in proportion to the value on the side. Emma rolls both dice. Find the probability that the sum of the outcomes will be odd.

2.1.29 Problem 29

Find the probability that the top two cards of a well-shuffled standard deck form a pair (two cards of the same rank).

2.1.30 Problem 30

5 fair coins are flipped. Given that you obtained 3 heads, find the probability that you obtained the sequence $THHTH$.

2.1.31 Problem 31

Consider a family with n children, where $n \geq 2$. Let X be the event that the family has at least one child of each gender, and let Y be the event that there is at most one girl in the family. Assuming that each child has an equal probability of being born a boy or a girl, determine the unique value of n such that events X and Y are independent.

2.1.32 Problem 32

A pack contains 6 red and 10 blue jellybeans. A child wants to eat 4 jellybeans and grabs into the pack to select 4 jellybeans one-by-one uniformly at random without replacement. Find the probability that the first two jellybeans are red and the last two are blue.

2.1.33 Problem 33

Someone leaked IP from a quant firm! A witness claims that the IP was leaked by a trader. Suppose that witnesses are correct $\frac{2}{3}$ of the time. At this particular firm, $\frac{2}{3}$ of the employees are traders, while the other $\frac{1}{3}$ are researchers. Taking into account the witness statement, what is the probability that the person who leaked IP was a trader?

2.1.34 Problem 34

A stoplight displays the colors red (stop), yellow (slow down), and green (go). The colors go in the sequence green $\rightarrow$ yellow $\rightarrow$ red. The light stays green for 40 seconds until it switches to yellow for 4 seconds. Afterwards, it turns red for 40 seconds before turning green again and repeating the cycle. Find the probability that the color of the stoplight switches in a 4 second interval.

2.1.35 Problem 35

Jeff is trapped in a room with 5 indistinguishable doors. Behind 2 of the doors are paths to freedom; one path is 2 miles long, and the other path is 5 miles long. Behind 3 of the doors are paths that magically lead back to the room; one path is 1 miles long, another path is 3 miles long, and the final path is 4 mile long. If Jeff returns to the room, the order of the doors are scrambled and the 5 doors are once again indistinguishable. What is the expected number of miles Jeff will travel before reaching freedom?

2.1.36 Problem 36

Two fair 6-sided dice are rolled and their upfaces are recorded. Find the probability that if both dice are rolled again, the values rolled on the second trial are the same as on the first trial.

2.1.37 Problem 37

You have 2 jars in front of you. Jar A has 3 blue and 4 red balls. Jar B has 6 blue and 4 red balls. If one ball is selected uniformly at random from each jar, what is the probability that the two selected balls match in color?

2.1.38 Problem 38

Meek Mill is performing at Spring Fair. The administration does not know where he is, and they estimate there is an 80% chance he is in Philadelphia and 20% chance he is in Baltimore. If he is in Baltimore, there is a 80% chance he will perform. If he is in Philadelphia, there is a 10% chance he will perform. Find the probability that if Meek Mill performs, he came from Philadelphia.

2.1.39 Problem 39

John has a standard deck of 52 cards. John identifies the ranks of the cards as $1 - 13$, where Ace is 1 and King is 13. He selects 4 cards uniformly at random from the deck. He then looks at the 4 cards he selected and removes the lowest ranked card from the subset of 4 and discards it. Afterwards, he selects another card uniformly at random from the remaining 48 card deck to replace the minimum. Find the expected value of the card John puts into the subset.

2.1.40 Problem 40

You are on a game show with 3 doors in front of you. One of the doors has \$1 million inside, while the other two are empty. In the final round, the host lets you spin a wheel that may reveal which door the \$1 million is in. This wheel will tell you the location of the \$1 million $\frac{3}{5}$ of the time. Otherwise, it tells you nothing, and you must guess uniformly at random. What is the probability you locate the \$1 million door?

2.1.41 Problem 41

You are on a game show with 3 doors in front of you. One of the doors has \$1 million inside, while the other two are empty. In the final round, the host lets you spin a wheel two times that may reveal which door the 1 million is in. On each spin, this wheel will tell you the location of the \$1 million with probability $3/5$, independent between spins. Otherwise, it tells you nothing about the location of the money. After two spins, if the wheel has not revealed to you the location of the 1 million, you must guess uniformly at random. What is the probability you locate the \$1 million door?

2.1.42 Problem 42

n coins are laid out in front of you. One of the coins is fair, while the other $n - 1$ have probability $0 < \lambda < 1$ of showing heads. If all n coins are flipped, find the probability of an even amount of heads.

2.1.43 Problem 43

Find the probability that a randomly selected integer from the set of positive divisors of 10^{99} is divisible by 10^{80} .

2.1.44 Problem 44

Let $N \sim \text{Poisson}(2)$. We flip a fair coin N times. Find the probability that no heads are obtained by the end of N flips. Your answer should be in the form e^a for a rational number a . Find a .

2.1.45 Problem 45

You are playing in a tennis tournament against Alice and Bob. In the tournament you will play 3 games, where if you win 2 consecutive games you win the tournament. The probability you beat Bob is better than beating Alice, let A represents Alice and B represents Bob. Answer 1 if you would prefer your three games to be ABA and 2 if you'd like your 3 games to be BAB .

2.1.46 Problem 46

Your older brother places 5 good candies and 5 bad candies in front of you, as well as two boxes. He asks you to put the candies into these two boxes, and he will, blindfolded, choose a random candy from a random box, which you get to keep. You strategize to put the candies into the boxes in such a way that maximizes your chances of receiving a good candy. With this strategy, what is the probability that you receive a good candy?

2.1.47 Problem 47

4 distinct integers are drawn from the set of the first 16 positive prime integers. Find the probability that the sum is even.

2.1.48 Problem 48

Garrett loves eating pizza, but enjoys savoring the slices too. The time it takes for him to eat a pizza slice is uniformly distributed between 1 and 5 minutes. If he has been eating for 2 minutes and still isn't done, find the probability that he finishes the slice in the next minute.

2.1.49 Problem 49

Drew is creating a playlist. There are 147 songs in the playlist, of which 4 are written by DaBaby and 6 are written by Rae Sremmurd. Drew randomly orders all of the songs in the playlist. Find the probability all of the Rae Sremmurd songs in the playlist come before the second DaBaby song.

2.1.50 Problem 50

There are 8 points on a circle. You draw all possible chords that can be formed by connecting each pair of points. Then, you randomly selects 4 chords. What is the probability that the 4 chords form a convex quadrilateral?

2.1.51 Problem 51

A poker hand consists of five cards drawn from a standard deck of 52 cards. Let p represent the probability of having a four-of-a-kind (i.e four of the five cards share the same rank). Calculate the reciprocal of p .

2.1.52 Problem 52

Angelina is playing a game which she wins with probability 0.1. She must pay 10 to play, and if she wins, she receives 80. If Angelina starts out with 30, to the nearest thousandth, what is the probability that she wins exactly once before losing it all?

2.1.53 Problem 53

45% of students don't use QuantEssential, 25% are heavy prep users and 30% are light prep users. If the heavy prep users are twice as likely to get an offer as light prep users, and light prep users are twice as likely to get an offer as those that don't use the platform, then what is the probability that if someone got an offer that they were a heavy prep user of QuantEssential?

2.1.54 Problem 54

You have a pile of 100 coins. 1 of the coins is an unfair coin and has heads on both sides. The remaining 99 coins are fair coins. You randomly select a coin from the pile and flip it 10 times. The coin lands heads all 10 times. Calculate the probability that the coin you selected is the unfair coin.

2.1.55 Problem 55

This weekend, there is a 40% chance it rains on Saturday and a 70% chance it rains on Sunday. Assuming independence, what is the probability it does not rain this weekend?

2.1.56 Problem 56

This weekend, the chance of rain on Saturday is 40%, and on Sunday, it's 70%. Without assuming independence, find the tightest upper limit on the probability that it doesn't rain at all this weekend.

2.1.57 Problem 57

There are three train cars: a blue one, a red one, and a green one. This train has a conductor whose job is to disconnect the back train cars and put it in front. They can do this as many or as few times as they want. If the three train cars are randomly ordered, what is the probability that the conductor can rearrange it into rainbow order? (Note: Here, rainbow order means red, then green, then blue).

2.1.58 Problem 58

3 red and n blue socks are in a drawer. We know that when two socks are drawn uniformly at random without replacement, the probability both socks are red is $\frac{1}{2}$. Find n .

2.1.59 Problem 59

Ronaldo lives in Portugal in a house numbered 122577. A hurricane hits Portugal, and two random numbers on Ronaldo's house fall off. After the hurricane passes, he sees the numbers on the ground, but has amnesia, so he doesn't remember what his house number is. So, he places the two numbers up in the two empty spots uniformly at random. Find the probability Ronaldo creates the number 122577 again.

2.1.60 Problem 60

What is the probability of flipping the same side of a coin 4 times a row?

2.1.61 Problem 61

Your dresser has two shelves, top and bottom. The top shelf contains gray socks with probability 0.4 and white socks with probability 0.6. The bottom shelf contains gray socks with probability 0.7 and white socks with probability 0.3. You pick a shelf at random. From that shelf, you pick 2 gray socks. What is the probability that you selected the top shelf? Assume you have so many socks that taking from either shelf does not change the probability of future socks that are drawn.

2.1.62 Problem 62

Billy is walking on a cartesian grid start at $(1, 2)$, she begins a 2D symmetric random walk. For making a decision on where to go, Billy has a $\frac{1}{4}$ chance each of going up, down, left, or right. There is a square formed with vertices $(0, 0)$, $(0, 4)$, $(4, 4)$, $(4, 0)$. Billy ends his random walk when he hits the square. What is the probability that Billy ends up on a vertical side of the square?

2.1.63 Problem 63

You are boating through the Mediterranean Sea when your motor breaks down. You remember learning from your divemaster that the probability of seeing at least one boat within a given hour is 73.7856%. Assume that boats are observed as a Poisson process with a fixed rate of observations per time step and that observations are independent in any mutually exclusive time period. What is the probability that you will see at least one boat within the next 10 minutes?

2.1.64 Problem 64

A tournament consistent of 32 teams plays in a round-robin style tournament. This means that every single team plays every other team exactly one time. Suppose that every team is equally skilled so that every team is equally likely to beat any other team in a given match. Each team keeps track of how many wins they have. Find the probability that every team has won a distinct number of games at the end of the tournament. The answer will be in the form $\frac{a!}{2^b}$ for integers a and b , where b is maximal. Find $a + b$.

2.1.65 Problem 65

Suppose we roll 5 standard fair dice and sum the upfaces of the largest 3 values showing. Find the probability that the sum is 18.

2.1.66 Problem 66

Four cards labelled water, earth, fire, and air are placed in front of you faced down. You flip each of the cards over one at a time. You win if you flip over both the air and water cards before the fire card. Otherwise, you lose. What is the probability that you win?

2.1.67 Problem 67

Three riflemen X, Y and Z take turns shooting at a target. X shoots first, Y second, and Z third, after which the cycle repeats with X again and again, until one of the riflemen hits the target. Each shot hits the target with probability 50%, independent of other shots. Find the probability X wins.

2.1.68 Problem 68

There are 5 babies in a room: 2 boys and 3 girls. one baby with unknown sex is added. One baby is random selected and it is a boy. What's the probability that the added baby is a boy? You may assume that each gender is equally likely at birth.

2.1.69 Problem 69

A 6-sided die shows the values 1 – 6 with respective probabilities $1/6, 1/5 - x, 1/10 + 2x, 1/5 - x, 1/6$, and $1/6$. Find the value of x that maximizes the probability of obtaining a sum of 7 when you roll this die twice.

2.1.70 Problem 70

When rolling two dice repeatedly, what is the probability of rolling a sum of 5 before rolling a sum of 2?

2.1.71 Problem 71

A letter is going to be sent to one of three mailboxes uniformly at random. These mailboxes are labelled 1 – 3. If the mail ends up in mailbox i , then there is a $\frac{i}{3}$ probability that the mail is actually read. Given that you checked Mailbox 1 and didn't read any letter, find the probability that the mail ended up in Mailbox 1.

2.2. Level 2

2.2.1 Problem 1

Cards are dealt one-by-one from a standard deck of 52 cards without replacement. The card dealing ends when either 2 kings appear OR at least 1 king and 1 ace appear, whichever comes first. Find the expected number of cards that are dealt in the game.

2.2.2 Problem 2

Ty, Guy, and Psy are all flipping fair until they respectively obtain their first heads. Let T , G , and P represent the number of flips needed for Ty, Guy, and Psy, respectively. Find $\mathbb{P}[T < G < P]$.

2.2.3 Problem 3

You and your opponent each draw a number from $U(0, 1)$ without revealing it. You each have the option to redraw and keep that value instead, and the other person doesn't know which choice they made. The person with the higher number wins. The optimal strategy is of the form reroll if the number is less than k . Find k to 3 decimal places.

2.2.4 Problem 4

A gambler starts with n , where $n \geq 1$ is an integer. Each round, the gambler wins or loses 1 with respective probabilities p and $1 - p$. Once the gambler has 0 left, he leaves the table. Find the probability that the gambler is ruined after $n + 2k$ rounds of playing the game, where $k \geq 0$ is an integer. The probability can be written as a function $f(p, n, k)$. Compute $f(1/3, 5, 3)$ to the nearest ten-thousandth.

2.2.5 Problem 5

Suppose you flip a coin that has probability $0 < p < 1$ of landing heads on each flip. You flip the coin until you have an equal amount of heads and tails appearing for the first time (of course, after the initial state of 0 for both). For any integer $n \geq 1$, find the probability that you flip the coin exactly $2n$ times. The probability can be written as a function $f(p, n)$. Compute $f(1/3, 5)$ to the nearest ten-thousandth.

2.2.6 Problem 6

You are in line at an ice cream shop when the manager announces that she will give a free sundae to the first person in line whose birthday is the same as someone who has already bought an ice cream. Assuming that you do not know anyone else's birthday and that all birthdays are uniformly distributed across the 365 days in a normal year, what position in line will you choose to maximize your probability of receiving the free sundae?

2.2.7 Problem 7

Assume you have n identical urns, each containing red and blue balls. The i th urn, where $1 \leq i \leq n$, contains 1 red ball and $2^i - 1$ blue balls. Suppose you randomly select an urn and select a ball from it, and it's red. Let $p(n)$ be the probability that if you replace this ball and again draw a ball at random from the same urn then the ball drawn on the second occasion is also red. Compute $\lim_{n \rightarrow \infty} p(n)$.

2.2.8 Problem 8

There are 10 parking spots in front of Quant Questions headquarters. By 9:00 AM, all 10 parking spots are occupied. The company's 3 executives head home early in the afternoon, vacating their parking spots at random times distributed independently and uniformly between 12:00 PM and 3:00 PM. Simon is one of Quant Questions' 9 lazy employees; lazy employees arrive at the office at random times, also distributed independently and uniformly between 12:00 PM and 3:00 PM. If a parking spot is vacant when a lazy employee arrives, then that employee will occupy that spot until 5:00 PM. Otherwise, the lazy employee will call in sick and return home. What is the probability that Simon calls in sick?

2.2.9 Problem 9

Subsets A and B are chosen uniformly at random from the collections of all subsets of a set X of cardinality 5. What is the probability that A is a subset of B ?

2.2.10 Problem 10

You're practicing soccer by taking 100 penalty kicks. Assume that you have made the first goal but missed the second. For each of the following kicks, the probability that you score is the fraction of goals you've made thus far. For example, if you made 17 goals out of the first 30 attempts, then the probability that you make the 31st goal is $\frac{17}{30}$. After 100 attempts, including the first two, what is the probability that you score exactly 66 penalty kicks?

2.2.11 Problem 11

Let the radius of a penny be 1. Assume that the thickness of the penny is non-negligible so that a flipped penny can land on its side. Find the thickness of the penny that, when flipped, has a $\frac{1}{3}$ chance of landing on its side. The answer is in the form $\frac{1}{\sqrt{k}}$ for an integer k . Find k .

2.2.12 Note:

There is no definite answer for this question. Take the approach of inscribing the penny in a sphere.

2.2.13 Problem 12

Two candidates, say A and B , are running for office. Candidate A received n votes, while Candidate B received m votes, with $n > m$. The $n + m$ votes are thrown into a box and shuffled around. Then, the votes are drawn without replacement one-by-one. A running tally of the number of votes for each candidate is kept. The probability that Candidate A is never behind in the voting count (excluding the initial state where both have 0) is a function $Q(n, m)$. Find $Q(100, 80)$.

2.2.14 Problem 13

Tim has 20 fair coins and Jeff has 22 fair coins. They both flip their coins. What is the probability that Jeff has more heads than Tim? Round to the nearest thousandth.

2.2.15 Problem 14

Two machines independently manufacture Bakugan and Beyblade toys. The time it takes each machine to produce a toy is $\text{Exp}(1)$ distributed. The manufacturing time is independent of all other toys. Assume there is no delay between when a machine produces a toy to when it starts the next toy. Find the probability that the third Bakugan is manufactured before the second Beyblade.

2.2.16 Problem 15

The surfaces of a $3 \times 3 \times 3$ cube (initially white) are painted red. The large cube is then cut up into 27 $1 \times 1 \times 1$ small cubes. One of the small cubes is selected uniformly at random and is rolled twice. The face appearing is red both times. What is the probability the cube selected was a corner cube?

2.2.17 Problem 16

Julie and Judy bet on a series of 8 successive coin tosses. After each toss, assuming that both players have at least \$1, Julie gives Judy \$1 if it is heads, and Judy gives Julie \$1 if it is tails. They both start out with \$4 each. Compute the probability that they break even.

2.2.18 Problem 17

You are playing a game with the casino. You and the dealer are each dealt a card from a fair, shuffled deck of 52 cards. If you have a strictly larger number than the dealer's, you win- else, you lose. What is the probability you win?

2.2.19 Problem 18

A chess tournament has 128 players, each with a distinct rating. Assume that the player with the higher rating always wins against a lower rated opponent and that the winner proceeds to the subsequent round. What is the probability that the highest rated and second-highest rated players will meet in the final?

2.2.20 Problem 19

A chess tournament has 128 players, each with a distinct rating. Assume that the player with the higher rating always wins against a lower rated opponent and that the winner proceeds to the subsequent round. Since the tournament's structure resembles that of a knockout bracket, 7 total rounds are played, including the final. What is the probability that the highest rated and the third-highest rated players will meet in the final?

2.2.21 Problem 20

Bill and Bob are simultaneously flipping coins with probability $1/3$ of heads per flip. Find the probability that Bill obtains his first heads in strictly less flips than Bob.

2.2.22 Problem 21

Ty and Guy are both flipping fair coins until the respectively obtain their first heads. Find the probability that it takes Ty at least 4 times as many flips to obtain his first heads as Guy.

2.2.23 Problem 22

Alice and Bob are playing a game where they toss a fair coin. The first person to flip two heads (not necessarily sequentially) wins the game. Alice starts and then gives the coin to Bob and the cycle repeats until one of them has flipped two heads. What is the probability that Alice wins this game?

2.2.24 Problem 23

You have two urns before you, X and Y . Urn X has 400 orange and 100 yellow balls. Urn Y has 100 orange and 400 yellow balls. 300 of the balls are randomly selected from Urn X and moved to Urn Y . Then uniformly at random, one ball in Urn B is selected from the 800 balls. What's the probability that this ball is orange?

2.2.25 Problem 24

Suppose that $X, Y \sim N(0, 1)$ IID. Find $\mathbb{P}[X > 0 \mid X + Y > 0]$.

2.2.26 Problem 25

Suppose that in the integer lattice $\mathbb{Z}^2$, each pair of vertices of Euclidean distance 1 has a path connecting them with probability p . Using the bound provided by sub-additivity, the interval of values of p so that the probability that the origin is in an infinite connected component is 0 is in the form $(0, d)$. Find d .

2.2.27 Problem 26

Suppose 6 points are chosen uniformly at random along the perimeter of the circle. The 6 points are then divided into three pairs, and a chord is drawn between every pair of points. What is the probability none of the chords intersect?

2.2.28 Problem 27

Jenny has a fair 6-sided die with numbers 1 – 6 on the sides. Jenny continually rolls the die and keeps track of the outcomes in the order they appear. Jenny rolls until she sees two 5s (not necessarily consecutive) OR both 4 and 6. Find the probability Jenny stops rolling due to seeing two 5s.

2.2.29 Problem 28

Suppose that Alice has m (6-sided dice) and Bob has n (8-sided dice). If S_m is the sum of the upfaces of all m of Alice's dice and T_n is the sum of the upfaces of all n of Bob's dice, find the product m^*n^* , where m^* and n^* are the smallest integer values of m and n such that $\mathbb{P}[S_{m^*} < T_{n^*}] = \mathbb{P}[S_{m^*} > T_{n^*}]$.

2.2.30 Problem 29

Noelle is preparing a tuna fish of length L in anticipation of dinner. She selects 2 cutting points uniformly at random and independently along the length of the fish so that the cutting points are on opposite sides of the midpoint of the fish. Find the probability that the distance between the two points Noelle selects is greater than $\frac{L}{3}$.

2.2.31 Problem 30

You're on a game show and are given the choice of three doors to choose from. Behind one door is a car and behind the other two are goats. You pick one of the doors at random and announce your choice to the game show host. The game show host, knowing which prize is behind each door, opens a door that you did not choose and shows a goat after hearing your initial choice. He offers you the opportunity to either keep your original door or switch to the other closed door. What is the probability that you win the car if you switch?

2.2.32 Problem 31

Suppose that German tanks are assigned distinct serial numbers $1, 2, \dots, N$. You observe 6 tanks with serial numbers 38, 17, 59, 42, 97, and 120. Under a frequentist approach, what is the best guess for N ?

2.2.33 Problem 32

We place the integers 1 – 9 (inclusive, no replacement) randomly on a 3×3 grid. What is the probability each row, column and diagonal add up to an odd number?

2.2.34 Problem 33

Let B_t be a standard Brownian Motion and define $T_a = \inf\{t > 0 : |W_t| > a\}$ for $a > 0$. Find $\mathbb{E}[e^{-\lambda T_a}]$ as a function of λ and a and evaluate with: ($\lambda = 4$ and $a = \ln(2)$)

2.2.35 Problem 34

A dealer is dealing a deck with 12 cards to 4 players. The deck consists of 4 Kings, 4 Queens, and 4 Jacks. What is the probability that every person receives the same hand of 1 of each card?

2.2.36 Problem 35

The dealer is known to deal cards from a deck that is comprised of 10 standard card decks. This means that there are 40 of each rank of card and 520 total cards. The dealer plays a game where he shuffles all 10 decks together randomly, and then he starts turning over cards from the top of the deck. Once he turns over 5 kings, the game is over. You win the game if you are closest to how many cards total (including the final king) the dealer turns over before the game ends. How many cards would you guess to

2.2.37 maximize your chances

?Not that this is different than the expected value.

2.2.38 Problem 36

Suppose you have 10 independent light bulbs labelled $1 - 10$ whose lifetime is modelled by $T_i \sim \text{Exp}(1/10)$ for each $1 \leq i \leq 10$. Furthermore, suppose you have an 11th light bulb whose lifetime is modelled by $X \sim \text{Exp}(1/90)$. Find $\mathbb{P}[X > T_1 + \cdots + T_{10}]$. The answer is in the form q^b , where $0 < q < 1$ is a rational number and b is an integer with b maximal. Find qb .

2.2.39 Problem 37

A frog starting at position 0 is going to jump on the integers. At each step, the frog will choose to jump forward 1 or 2 steps with equal probability. Let p_k be the probability that the frog hits position $k > 0$. Find the largest value of p_k .

2.2.40 Problem 38

Jake has 200 black, 400 white, and 600 green marbles in a container. He draws the marbles one by one without putting any back. What is the probability that at least 1 white and 1 green marble remain in the container right after the last black marble is drawn?

2.2.41 Problem 39

Three fair 6-sided dice are rolled and their upfaces are recorded. Find the probability that the values showing upon rolling all three dice again is the same as the original three values recorded.

2.2.42 Problem 40

You flip 100 fair coins simultaneously. What is the probability that you observe an odd number of heads?

2.2.43 Problem 41

Suppose we have the values $1 - 6$ in a bowl. We draw them without replacement, noting the order in which we selected them. We multiply the first two values together, the next two values together, and the last two values together. Lastly, we add the three products above. Find the probability the sum is odd.

2.2.44 Problem 42

Consider a particle that performs a random walk on the integers starting at position 0. At each step, the particle moves from position i to position $i + 1$ with probability p , while the probability it moves from i to $i - 1$ is $1 - p$. If $p = 1/3$, find the probability the particle ever reaches position 1.

2.2.45 Problem 43

Consider a particle that performs a random walk on the integers starting at position 0. At each step, the particle moves from position i to position $i + 1$ with probability p , while the probability it moves from i to $i - 1$ is $1 - p$. If $p = 1/3$, find the expected number of steps until the particle reaches 1. If the answer is infinite, answer -1 .

2.2.46 Problem 44

You and 4 other people are sitting in a circle. You are given a ball to start the game. Every second of this game, the person with the ball has three choices they can make. They can either pass the ball to the left, pass the ball to the right, or keep the ball (all with equal probability). This game goes on till someone keeps the ball. What is the probability that you are the person to end the game and keep the ball?

2.2.47 Problem 45

Suppose that X and Y are two IID standard normal random variables. Compute $\mathbb{P}[Y > \sqrt{3}X \mid Y > 0]$.

2.2.48 Problem 46

Suppose $N > 0$ is an integer. Let X_1, X_2, X_3 be independently selected uniformly at random from the set $\{0, 1, \dots, N\}$. Let $p_N = \mathbb{P}[X_1 + X_2 + X_3 = N]$. Find $\lim_{N \rightarrow \infty} Np_N$. If this limit does not exist, enter -1 .

2.2.49 Problem 47

An urn contains 20 balls colored each of the 7 colors of the rainbow (140 total balls). We select balls one-by-one without replacement. Given that in the first 70 draws we selected 5 more red balls than yellow, find the probability the 71st ball drawn is yellow.

2.2.50 Problem 48

25 fair coins are lined up and flipped once. Afterwards, remove all of the coins that showed tails, and repeat the aforementioned process until you have no coins left or all of the coins show up heads. Find the probability that you end the game by having all of the coins show heads.

2.2.51 Problem 49

You and your friend are bored one day and decide to play Russian Roulette. The six-chambered revolver is loaded with one bullet. To start the game the cylinder is spun to randomize the order of the chambers. The two of you take turns pulling the trigger until the person who fires the gun loses. Considering the cylinder is not re-spun after each turn, what is the probability that you win if your friend goes first?

2.2.52 Problem 50

You and your friend decide to play a game of Russian Roulette. The gun is a six-chambered revolver and its loaded with one bullet. To start each round, the cylinder is spun to randomize the order of the chambers. The two of you take turns pulling the trigger until the person who fires the gun loses. Given that the cylinder is re-spun after each turn, what is the probability that you win if your friend goes first?

2.2.53 Problem 51

Consider n people $P_1, \dots, P_n$. P_1 receives binary information "yes" or "no" and will pass this information along to P_2 . More generally, P_i will pass information along to P_{i+1} . However, P_i will transfer the information that they hear with probability p and transfer the opposite information with probability $1 - p$, where $0 < p < 1$, independent of all other people. Let A_i be the event that P_i transfers the original information (i.e. what P_1 received) to P_{i+1} , and let the probability of this be p_i . Compute $\lim_{n \rightarrow \infty} p_n$. If this limit does not exist, answer -1 .

2.2.54 Problem 52

Roll a fair standard 6-sided die until a 6 appears. Given that the first 6 occurs before the first 5, find the expected number of times the die was rolled.

2.2.55 Problem 53

A 6-sided die with values 1–6 is weighted such that the probability of rolling the values 2, 3, 4, 5, and 6 are the same whether we roll this die once or if we roll the die twice and take the sum of the two outcomes. Let p_1 be the probability that the value 1 is obtained on this die. p_1 is the solitary positive solution to the polynomial equation $\sum_{k=1}^6 c_k p_1^k = 1$ for some real constants $c_1, \dots, c_6$. Find $\sum_{k=1}^6 c_k$.

2.2.56 Problem 54

In a drawer, Sandy has 5 pairs of socks, each pair a different color. On Monday, Sandy selects two individual socks at random from the 10 socks in the drawer. On Tuesday, Sandy selects two of the remaining 8 socks at random, and on Wednesday, two of the remaining 6 socks at random. Find the probability that Wednesday is the first day Sandy selects matching socks.

2.2.57 Problem 55

A square of side length 20 is formed in front of you. You select a point uniformly at random from the interior of the square. You then proceed to form a circle of radius $R \sim \text{Unif}(0, 10)$, independent of the point selected. Find the probability that the circle you create does not intersect the square at any point.

2.2.58 Problem 56

Suppose Jimmy and Simon are selecting uniformly random numbers. Jimmy selects from the set $\{1, 2, \dots, 1000\}$ and Simon selects from $\{1, 2, \dots, 3000\}$. Compute the probability that Simon chooses a strictly larger number than Jimmy

2.2.59 Problem 57

A point P is chosen within triangle $\triangle ABC$. Compute the probability that the area of $\triangle ABP$, $\triangle BCP$, or $\triangle CAP$ is no more than one fourth that of $\triangle ABC$.

2.2.60 Problem 58

Bob proposes a game where Alice and Bob are each given a coin. Each is allowed to decide the probability of heads for their coin. Afterwards, they both flip their coins. If it comes up HH , Alice gives Bob \$6. If it comes up TT , Alice gives Bob \$4. Otherwise, Bob gives Alice \$5. However, Alice must tell Bob the success probability p_1 they have chosen before Bob decides his p_2 . The interval of values for p_1 such that, regardless of what Bob selects, Bob has non-positive expected value in the game is in the form $[a, b]$, where a and b are rational numbers. Find $b - a$.

2.2.61 Problem 59

An ant starts at the origin in the plane. At each step, with probability $\frac{1}{4}$, the ant will move one unit north, south, east, or west. What is the probability that the ant returns to the origin before hitting any point on the boundary of the square with vertices at $(\pm 2, \pm 2)$?

3. Discrete Distributions and Moments

3.1. Level 1

3.1.1 Problem 1

Two identical sets of cards are in front of you. Each set has 5 cards labeled $0 - 4$. You pick a card from each set and add them together. What is the standard deviation of sum of the cards chosen?

3.1.2 Problem 2

Brad and Chad both flip coins with probability p of heads on each flip. Let B be the event that it takes strictly less flips for Brad to get his first heads than Chad, C be the event that it takes Chad strictly less flips than Brad to get his first heads than Brad, and E be the event that it takes them an equal number of flips to obtain their first heads. Find the value of p such that $\mathbb{P}[E] = \mathbb{P}[C]$.

3.1.3 Problem 3

Let F_n be the number of fixed points of a random function $f : S_n \rightarrow S_n$, where $S_n = \{1, 2, \dots, n\}$. Find $\lim_{n \rightarrow \infty} \mathbb{P}[F_n = 5]$. The answer is in the form $\frac{c}{e}$, where e is Euler's constant and c is a constant. Find c .

3.1.4 Problem 4

Suppose that two integers x and y are uniformly at random selected from $S = \{-10, -9, \dots, 9, 10\}$. Find the probability that $\max\{0, x\} = \min\{0, y\}$.

3.1.5 Problem 5

3 people flip a weighted coin with probability of heads $p = 0.25$ until they obtain their first tails. Find the probability that none of the three people flip more than twice.

3.1.6 Problem 6

Gabe makes rugs. The number of imperfections follows a Poisson distribution with an average of 4 per square yard. The probability that a 3-square-yard rug sample contains at least 1 imperfection can be expressed in the form $1 + \frac{a}{e^b}$. Compute $a + b$.

3.2. Level 2

3.2.1 Problem 1

A fair 6-sided die is rolled repetitively, forming a sequence of values, under the following rules: If any even value is rolled, add it to the current sequence. If a 3 or 5 is rolled, discard the entire sequence and don't add the 3 or 5 to the start of the new sequence. If a 1 is rolled, add the 1 to the current sequence and end the game. Find the expected length of the sequence at the end of the game.

For example if the sequence of rolls is 34265241, we would reset to an empty sequence at the first roll, reset to an empty sequence at the 5, and our final sequence would be 241, which is of length 3.

3.2.2 Problem 2

You're at the bank on a lucky day. The banker will randomly choose a positive integer, and so will you. Both of you can set a probability distribution over the positive integers that you select from. If you both select the same integer, n , you win n^2 dollars; otherwise, you get nothing. Neither party knows the other's distribution. Assuming optimal strategies from both sides, what is your expected payout? Express your answer in the form $\frac{a}{\pi^b}$ for integers a and b , and find the product ab .

3.2.3 Problem 3

Suppose that you roll a standard fair 6-sided die n times. Let X be the random variable representing the number of times you roll a 1 and Y be the random variable representing the number of times you roll a 5. Calculate $\text{Corr}(X, Y)$.

4. Continuous Distributions and Moments

4.1. Level 1

4.1.1 Problem 1

A fair coin is tossed 100 times. Using the Central Limit Theorem without continuity correction, estimate the probability that at least 60 heads are observed. Round to 5 digits after the decimal point.

4.1.2 Problem 2

Let $X_1, \dots, X_{324}, Y_1, \dots, Y_{324} \sim \text{Beta}(1, 2)$ IID. Define $S_{324} = X_1 + \dots + X_{324}$ and $T_{324} = Y_1 + \dots + Y_{324}$. Approximate $\mathbb{P}[S_{324} - T_{324} > 10]$. Your answer should be in the form $\Phi(a)$, where Φ is the standard normal CDF. Find a .

4.1.3 Problem 3

A bomb lands uniformly at random within a circle with radius 1 mile such that the target is at the center of the circle. When the bomb explodes, it wipes out everything within $\frac{1}{2}$ miles of its landing spot. What is the probability that the target remains unharmed.

4.1.4 Problem 4

Gabe has 2 loaves of bread of lengths 5 and 8. He has no control of where he cuts the length 8 bread piece, so he cuts it at a uniformly random position along its length. What is the probability that the loaf of length 5 and the two resulting pieces of bread from the cut form a triangle?

4.1.5 Problem 5

Jon loves cheese. He decides to make 100 blocks of cheese. The distribution of the weight (in grams) of each block he makes follows IID $\text{Exp}\left(\frac{1}{250}\right)$ distribution. Let W_i denote the weight of the i th block of cheese, and T_{100} represent the total weight of the 100 blocks of cheese. Using Markov's Inequality, what is an upper bound on $\mathbb{P}[T_{100} > 26000]$?

4.1.6 Problem 6

$Z \sim \mathcal{N}(0, 1)$, $X = Z$, $Y = Z^2$. Compute $\text{Cov}(X, Y)$.

4.1.7 Problem 7

Suppose that a point (X, Y, Z) is uniformly randomly selected from the upwards-facing cylinder of radius r and height h whose lower base is centered at $(0, 0, 0)$. Find the probability that this point also lies in the cone whose base and height is the same as the cylinder.

4.1.8 Problem 8

X is a Gaussian random variable with $\mu = 50$, $\sigma^2 = 4$. what is $\mathbb{P}(X > 54)$? Round to the nearest thousandth.

4.1.9 Problem 9

Let $X \sim \text{Unif}(0, \lambda)$ and $Y \sim \text{Exp}(\lambda)$, where $\lambda > 0$ is a constant. For what value of λ do X and Y have the same mean? Find λ^2

4.1.10 Problem 10

An ellipse is centered at the origin. Let A and B be the horizontal and vertical radii lengths of the ellipse, respectively. Assume that $A, B \sim \text{Exp}(1)$ and are independent. Recall that the area of an ellipse with radii a and b is given by πab . A circle with diameter A is created as well. Find the probability the area of the circle is larger than the area of the ellipse.

4.1.11 Problem 11

Compute $\frac{\Gamma\left(\frac{9}{4}\right)}{\Gamma\left(\frac{1}{4}\right)}$.

4.1.12 Problem 12

Let $X \sim \text{Gamma}(2, 4)$. Note that this is the shape-scale parameterization of Gamma. Evaluate $\mathbb{E}[X^6]$. The answer is in the form $a^b \cdot c!$ for integers a, b, c , with a minimal. Find $a + b + c$.

4.1.13 Problem 13

Mordecai has a large target of radius 10 and a laser pointer. He is clumsy, so he points the laser at a uniformly random point on the target. Let D represent the random distance from the laser to the center of the target. He offers you to play the following game: Mordecai points the laser at the target one time. Before doing so, you fix a value $0 \leq r \leq 10$. If your laser is within distance r away from the center, you must pay Mordecai $\$(10 - r)$. Otherwise, Mordecai pays you $\$r$. Find the value of r that maximizes your expected winnings from this game.

4.1.14 Problem 14

Two points are uniformly at random selected between -1 and 2 . What is the probability that the origin lies between those two points?

4.1.15 Problem 15

A right triangle is being formed with legs labeled A and B . The random lengths of legs A and B are both IID $\text{Unif}(0, 1)$ RVs. Let θ be the angle (of random degree measure) that is opposite of side A . Find the probability $\theta > \frac{\pi}{3}$. The answer is in the form $\frac{a}{b\sqrt{c}}$ for integers a, b, c that are relatively prime. Find abc .

4.1.16 Problem 16

Let $X_1, X_2, X_3, X_4 \sim \text{Unif}(0, 4)$ IID. Find the probability that the maximum of these 4 random variables is in the interval $(2, 3]$.

4.1.17 Problem 17

A snowman is created by placing three spherical snowballs of radii 8, 11, and 14 inches on top of one another in descending size. Assume that every single spot of each sphere is visible on the snowman, you uniformly pick a random point on the surface of the snowman. Find the probability that this point is at most 39 inches above the ground.

4.1.18 Problem 18

Liverpool FC sells player jerseys at random prices. The price of a Mohamed Salah jersey is normally distributed with mean \$80 and standard deviation \$15. The price of a Sadio Mane jersey is normally distributed with mean \$70 and standard deviation \$20, independent of Salah jerseys. Gabe wants to be 97.5% sure that he can purchase both jerseys. How much money (in dollars) should Gabe bring with him? For the purposes of this question, take $\Phi^{-1}(0.95) = 1.65$ and $\Phi^{-1}(0.975) = 1.96$

4.1.19 Problem 19

Let $X_1, X_2, \dots, X_{20}$ be IID $\text{Unif}(0, 1)$ random variables. Compute $\mathbb{P}[X_1 > \max\{X_2, X_{10}, X_{15}\}]$

4.1.20 Problem 20

Let $X, Y \sim \text{Unif}(0, 1)$ IID. Find $\mathbb{P}\left[XY > \frac{1}{2}\right]$. Round your answer to the nearest hundredth.

4.1.21 Problem 21

Let $X \sim \text{Unif}(0, 1)$. Find $\text{Var}(X)$.

4.1.22 Problem 22

Let $X, Y, Z \sim N(12, 12)$ IID. Compute $\mathbb{P}[X - Y > Z]$. The answer is in the form $\Phi(a)$ for some a . Find a .

4.2. Level 2

4.2.1 Problem 1

Ty, Guy, and Psy are all flipping fair coins until they respectively obtain their first heads. Let T , G , and P represent the number of flips needed for Ty, Guy, and Psy, respectively. Find $\mathbb{P}[T \leq G \leq P]$.

4.2.2 Problem 2

Let $X_1, X_2, \dots \sim \text{Unif}(0, 1)$ IID. Let N be the first index n where $X_n \neq \min\{X_1, \dots, X_n\}$. Find $\mathbb{E}[X_{N-1}]$ i.e. the smallest value among the first N values selected. The answer will be in the form $a + be$ for integers a and b . Note here that e is Euler's constant. Find $a + b$.

4.2.3 Problem 3

A stick of 1 metre is randomly broken into three pieces. Given each break follows a uniform distribution along the stick, what is the probability that the three segments can form a valid triangle?

4.2.4 Problem 4

A lumberjack cuts a piece of wood of 1 meter in length at 2 uniformly randomly selected locations along the length of the wood. Find the probability the shortest piece of wood is at most 5 centimeters.

4.2.5 Problem 5

You sample 101 uniformly random numbers in the interval $(0, 1)$. Find the probability that no two of the values selected are within a distance of $\frac{1}{1000}$ of one another. The answer should be in the fully reduced form $\left(\frac{a}{b}\right)^c$. Find $a + b + c$.

4.2.6 Problem 6

Suppose that $X, Y \sim \text{Unif}(0, 1)$ IID. Compute the probability that $\lceil \frac{Y}{X} \rceil$ is a perfect square. The answer is in the form $q - \frac{\pi^2}{a}$ for a rational q and an integer a . Find aq .

4.2.7 Problem 7

We make a table from a circular disk and three legs. We attach the three legs to the circumference of the circular disk. What is the probability that the table stands up?

4.2.8 Problem 8

Suppose that we generate $X_1, X_2, \dots, X_n \sim \text{Unif}(0, 1)$ IID. Let $U = \min\{X_1, \dots, X_n\}$ and $V = \max\{X_1, \dots, X_n\}$. Compute $\mathbb{P}[U + V > 1]$.

4.2.9 Problem 9

Let (X, Y) follow a Bivariate Normal distribution with X and Y marginally standard normal and $\text{Corr}(X, Y) = \rho$. Compute $\text{Cov}(X, Y^2)$.

4.2.10 Problem 10

Suppose you continually randomly sample nested intervals from $[0, 1]$, halving the size each time. That is, the next interval is $[x, x + 0.5]$, where $x \sim U(0, 0.5)$, and so on. What is the variance of the point this converges to?

4.2.11 Problem 11

Consider the following joint pdf:

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} c(1 - x_2), & 0 \leq x_1 \leq x_2 \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

where c is a constant such that f_{X_1, X_2} is a valid joint pdf. Compute $\text{Cov}(X_1, X_2)$.

4.2.12 Problem 12

Let $X \sim \text{Exp}(1)$ and $Y \mid X = x \sim \text{LogNorm}(0, x)$. Find $\text{Cov}(X, Y)$. Note that we say a random variable $R \sim \text{LogNorm}(\mu, \sigma^2)$ if $\log(R) \sim N(\mu, \sigma^2)$.

4.2.13 Problem 13

Let $X_1, X_2, \dots$ be IID $\text{Exp}(1)$ random variables and $M_n = \max\{X_1, \dots, X_n\}$. Define p_n as $\mathbb{P}[M_n \leq \ln(n)]$. Evaluate $\lim_{n \rightarrow \infty} p_n$. Given your answer should be in the form $\frac{b}{e}$, where b is some constant and e is Euler's constant, what is b ?

4.2.14 Problem 14

Mario and Luigi are throwing Fireballs. Each one throws a fireball twice. Let M_1 and M_2 represent the distance of Mario's 2 attempts, while L_1 and L_2 represent the distance of Luigi's 2 attempts. We have that $M_i \sim \text{Exp}(1)$ while $L_i \sim \text{Exp}(2)$. All attempts are independent of all other attempts. Let $M = \max\{M_1, M_2\}$ and $L = \max\{L_1, L_2\}$. Find $\mathbb{P}[M > 2L]$.

4.2.15 Problem 15

Suppose that $X \sim \text{Gamma}\left(8, \frac{1}{3}\right)$. Compute $\text{Var}(e^X)$. The answer is in the form $a^b - \left(\frac{a}{2}\right)^c$ for integers a, b , and c . Find abc .

4.2.16 Problem 16

Let $X, Y \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$. Compute the probability that $\lfloor \log_2 X \rfloor = \lfloor \log_2 Y \rfloor$.

4.2.17 Problem 17

Suppose $X_1, X_2 \sim \text{Exp}(1)$ IID. Let O_1 and O_2 be the minimum and maximum order statistics, respectively. Find the probability that the range of X_1 and X_2 is larger than the midpoint of X_1 and X_2 .

4.2.18 Problem 18

Jimmy picks a number uniformly at random from $(0, 1)$. If Jimmy chooses x , then Jon picks a number from $(x, 1)$ uniformly at random. If Y represents the number Jon selects, in simplest form, find $\frac{\mathbb{E}[Y]}{\text{Var}(Y)}$.

4.2.19 Problem 19

Suppose that $X_1, X_2, X_3, X_4 \sim \text{Unif}(0, 1)$ IID. Draw two line segments: One between X_1 and X_2 and one between X_3 and X_4 . Find the probability that there is overlap between the two line segments.

4.2.20 Problem 20

Gabe and a student have decided to meet up to discuss probability. The student is very prompt and will show up at a uniformly random time between 4 and 5 PM. Gabe is a tad late, so he will show up at a uniformly random time between 4 : 30 and 5 PM. For whichever person gets there first, they will wait up to 10 minutes, and if the other person has not shown up, they will leave. Find the probability that the meeting occurs.

4.2.21 Problem 21

$X \sim \text{Unif}([0, \pi])$. Find $\cos(\mathbb{E}[X | \sin(X)])$.

4.2.22 Problem 22

Let X be a continuous random variable with PDF $f(x) = \frac{\lambda}{2}e^{-\lambda|x-\mu|}$ for all some real constant λ and $\mu > 0$. Given that μ_X and σ_X are the mean and standard deviation of X , compute the skewness of X i.e. $\mathbb{E}\left[\left(\frac{X - \mu_X}{\sigma_X}\right)^3\right]$, You may assume without proof the first three moments of X are finite.

4.2.23 Problem 23

Let $X, Y \sim N(0, 1)$ IID. Compute $\mathbb{E}[X | X^2 + Y^2]$.

4.2.24 Problem 24

Let $X, Y \sim \text{Unif}(0, 1)$ IID. Compute $\mathbb{P}[X + Y \geq 2|X - Y|]$.

4.2.25 Problem 25

Let $X, Y, Z \sim \text{Unif}(0, 1)$ IID. Find the probability that a triangle of side lengths X, Y , and Z can be formed.

5. Multivariate Distributions and Moments

5.1. Level 1

5.1.1 Problem 1

8 cards are dealt out from a standard deck. Find the variance of the number of hearts in the 8 cards.

5.1.2 Problem 2

A player rolls a standard fair 6-sided die. If the player rolls a 6, the game ends and the player receives no payout. Otherwise, the player may quit and receive $\$k$, where k is the upface of the previous roll. Otherwise, the player may roll again under the same rules. Assuming optimal play from the player, what is their expected payout?

5.1.3 Problem 3

You are given a fair 6-sided die and play the following game: You receive the value of the face-up side on each roll. If you roll an odd number, the game ends. If you roll an even number, you flip a fair coin. If the coin lands on tails, the game ends. If it lands on heads, you roll the die again and repeat the coin flip process. How much do you expect to make from this game?

5.1.4 Problem 4

You are given the following information:

$$\mathbb{E}[X_1] = -3$$

$$\mathbb{E}[X_2] = 7$$

$$\text{Var}(X_1) = 24$$

$$\text{Var}(X_2) = 6$$

Compute the maximum possible value for $\text{Cov}(X_1, X_2)$.

5.1.5 Problem 5

X_1 and X_2 have a correlation coefficient $\rho_{X_1, X_2} = 0.2$. $Y_1 = 1 + 2X_1$, and $Y_2 = 3 - 4X_2$. Compute the correlation coefficient for Y_1 and Y_2 which is ρ_{Y_1, Y_2}

5.1.6 Problem 6

Suppose that $Z \sim N(0, 1)$ and $Y \sim \chi^2(u)$ are independent. Define $W = Z/\sqrt{Y}$. Compute $\text{Cov}(Y, W)$.

5.1.7 Problem 7

A fair 6-sided die is rolled until the sum of all upfaces is even. Find the expected number of rolls performed.

5.1.8 Problem 8

You roll a fair die until a value other than 1 appears and are paid the amount on the die that appears on the last roll. What is the fair value of this game?

5.1.9 Problem 9

Suppose X and Y are two random variables with respective variances 9 and 16. Find the standard deviation of $X + Y$ if $\rho(X, Y) = -3/8$.

5.1.10 Problem 10

Alice and Bob start with \$15 and \$10, respectively. They bet \$1 at a time on opposite outcomes of a fair coin. If Alice goes bankrupt, she has the ability to buy in again with \$15, while Bob keeps the stack he has won. If she goes bankrupt again, she loses the game. Find the probability Bob loses the game.

5.1.11 Problem 11

Let X, Y , and Z be uncorrelated random variables with respective variances 64, 36, and 225. Let $U = X + Y$ and $V = X + Z$. Find $\text{Corr}(U, V)$.

5.1.12 Problem 12

Let B_1, B_2 , and B_3 be standard Brownian Motions. Find $\text{Cov}(B_1(1)+B_2(1), B_2(2)+B_3(2))$.

5.2. Level 2

5.2.1 Problem 1

Let W_t be a standard Brownian Motion. We define the Brownian Bridge on $[0, 1]$ as $X_t = W_t - tW_1$. Find $\mathbb{E}[X_{1/2} \mid X_{3/4} = 3]$.

5.2.2 Problem 2

Suppose you play a game where you continually roll a die until you obtain either a 5 or a 6. If you receive a 5, then you cash out the sum of all of your previous rolls (excluding the 5). If you receive a 6, then you receive no payout. You do not have the decision to cash out mid-game. What is your expected payout?

5.2.3 Problem 3

Suppose you play a game where you continually roll a die until you obtain either a 5 or a 6. If you receive a 5, then you cash out the sum of all of your previous rolls (excluding the 5). If you receive a 6, then you receive no payout. You also have the decision to cash out mid-game for the sum of all your previously obtained rolls. Assuming optimal play, what is your expected payout? Round your answer to the nearest hundredth.

5.2.4 Problem 4

You are in a circle with 100 points labelled 0 – 99 clockwise. You start at 1. At each turn, you move one unit left or right with equal probability. What is the expected time to hit 0?

5.2.5 Problem 5

You have two fair 6-sided dice. At each round, you roll both dice and record the upfaces. Find the expected number of rounds you need to perform to observe all 6 faces of the dice.

5.2.6 Problem 6

Bobby draws from a deck of cards repeatedly, with replacement. Given the deck is a standard 52 card deck, what is the expected number of draws to get 4 jacks in a row?

5.2.7 Problem 7

Suppose that $X_1, \dots, X_{150}$ are all IID random variables with variance 1. Define $U = 3(X_1 + \dots + X_{50}) + X_{51} + \dots + X_{100}$ and $V = X_{51} + \dots + X_{100} + 3(X_{101} + \dots + X_{150})$. Find $\text{Corr}(U, V)$.

5.2.8 Problem 8

Suppose that X and Y are perfectly correlated i.e. $\rho(X, Y) = 1$ with $\sigma_Y > \sigma_X$. Find the correlation of $X + Y$ and $X - Y$.

5.2.9 Problem 9

Suppose that X and Y are perfectly correlated i.e. $\rho(X, Y) = 1$ with $\sigma_Y < \sigma_X$. Find the correlation of $X + Y$ and $X - Y$.

5.2.10 Problem 10

Suppose that X and Y are perfectly negatively correlated i.e. $\rho(X, Y) = -1$ with $\sigma_Y > \sigma_X$. Find the correlation of $X + Y$ and $X - Y$.

5.2.11 Problem 11

A frog has 100 lily pads arranged in a circle. Each one has a distinct number $0, 1, \dots, 99$ on it, arranged in increasing order counter-clockwise starting from 0. The frog starts at lily pad 0 and hops to the closest lily pad on the left or right of its current position with equal probability per hop. Find the probability that when the frog lands on lily pad 50 for the first time, it has visited every other lily pad not labeled 50.

6. Random Variables and Expectation

6.1. Level 1

6.1.1 Problem 1

Anna and Brenda are playing a game. They repeatedly toss a coin. Anna wins if 3 heads appear. Brenda wins if 3 tails appear. The heads and tails do not need to be consecutive. What is the expected number of coin tosses for a winner to be determined?

6.1.2 Problem 2

Let X and Y be the upfaces of two independent rolls of fair 30-sided dice with values $1 - 30$ on the sides. Find $\mathbb{E}[|X - Y|]$.

6.1.3 Problem 3

Find the expected number of flips needed to observe 3 heads on a coin with probability $\frac{1}{4}$ of heads per flip.

6.1.4 Problem 4

A casino offers you a game with a 6-sided die where you are paid the value of the roll. The casino lets you roll the first time. If you are happy with your roll, you can cash out. Else, you can choose to re-roll and cash out on this second value. What is the fair value of this game?

6.1.5 Problem 5

Alice rolls a fair 6-sided die with the values 1 – 6 on the sides. She sees that value showing up and then is allowed to decide whether or not she wants to roll again. The re-roll costs \$1. If she decides to stop, Alice receives a payout equal to the upface of the die. If she rolls again, she receives a payout equal to the upfaces of the second roll. Assuming Alice plays optimally and can only re-roll at most once, what is her expected payout on this game?

6.1.6 Problem 6

There are four balls, two black and two yellow, in a box. You pick out one at a time at random without replacement. Before picking each one out, you guess at the color, and if you're right, you receive in dollars the number of balls left in the container before you chose it. For example, if you guess the first ball correctly, you would receive \$4 because there were 4 balls remaining in the box before you picked one out. Assuming optimal play, what is the expected payoff for this game?

6.1.7 Problem 7

You are anticipating the arrival of a bus to reach an interview on time! The bus operates on a consistent schedule, arriving every 10 minutes. If you show up at a randomly chosen time during the day, what is the expected duration until the next bus arrives (in minutes)?

6.1.8 Problem 8

Fred is selling his old car. He will sell it to the first bidder that places a bid of at least \$9000. He receives bids for the car that are all independent and identically distributed exponential random variables with mean \$5000. Find the expected number of bids that Fred receives before selling his car. This is inclusive of the bid that he receives that makes him sell the car. The answer is in the form e^a for a rational number a . Find a .

6.1.9 Problem 9

You play a game where you initially start with \$1. For each roll of a die where you don't obtain a 1, you will multiply your current payout by α . At any point you are allowed to stop the game and take your payout. Otherwise, if you don't stop and roll a 1, you receive no payout. Find the value of α such that your expected payout on the game is constant regardless of the number of times you roll.

6.1.10 Problem 10

You toss a fair coin 100 times and record the outcomes. How many runs will you have on average? A run is classified as the longest continuous flips of heads or tails.

6.1.11 Problem 11

Find the expected number of flips of a fair coin needed to either have 10 more heads flipped than tails or 7 more tails flipped than heads.

6.1.12 Problem 12

Your company has a 20% chance of being acquired between Monday evening and Tuesday morning. On Monday, one share is worth \$30, though if acquired there is a 50% chance that one share of stock is worth \$50 and a 50% chance that one share of its stock is worth $\$x$. If not acquired, there is a 60% chance that one share of its stock is worth \$0 and a 40% chance that one share of stock is worth \$10. A fair contract that awards the buyer the right but not the obligation to buy the stock at \$40 on Tuesday is worth \$15. Compute x .

6.1.13 Problem 13

9 married couples are standing in a line. They mix around and form 9 pairs uniformly at random. Find the expected number of couples that are paired up together. Note that pairs can also consist of members of the same gender (there are 9 males and 9 females)

6.1.14 Problem 14

Two fair coins are flipped at once. You receive \$2 if exactly one heads that appears, but you lose \$7 if you flip two heads. What is your expected profit/loss on this game if you play once?

6.1.15 Problem 15

You have a circular dartboard of radius 1. You throw a dart such that it lands uniformly at random on the surface of the dartboard. Find the expected radial distance that the dart lands from the center.

6.1.16 Problem 16

A box consists of 1 \$100 bill, 3 \$20 bills, and 4 \$10 bills. Given you select 7 bills out of this bin uniformly at random without replacement, find the expected value of the sum of all the bills you select.

6.1.17 Problem 17

Suppose we are flipping a fair coin 5 times. Find the expected number of heads we obtain given that the first and last flips are of opposite parity (i.e. one is heads and one is tails).

6.1.18 Problem 18

Gabe has a crippling gambling addiction. As such, he plays a game where the winning number on the n th game, where $n \geq 1$, is uniformly at random selected from the set $\{1, 2, \dots, n + 2\}$. Let W be the random variable representing how many times Gabe plays before he wins. Find $\mathbb{E}[W]$. If the answer is infinite, answer -1 .

6.1.19 Problem 19

Suppose that you generate a uniformly random number in the interval $(0, 1)$. You can either keep this number or generate a new number once. Your payout is the last number generated. Find your expected payout under a rational strategy.

6.1.20 Problem 20

Instead of the Moment Generating Function (MGF), for non-negative integer-valued discrete random variables X , we can define the *Probability Generating Function* (PGF), which is given by $p(z) = \mathbb{E}[z^X]$ for $|z| \leq 1$. Find the PGF of $X \sim \text{Geom}(p)$. Evaluate this function for $z = \frac{1}{3}$ and $p = \frac{2}{3}$.

6.1.21 Problem 21

A farmer wants one male and one female goat for breeding. He goes to Mount Everest to find goats. Each goat is known to be male with probability $\frac{3}{4}$, independent of all other goats. The farmer takes a goat, examines it, and then leaves with the first two goats that are of opposite genders. If N is the total number of goats that the farmer examines until he observes two goats of opposite genders, find $\mathbb{E}[N]$.

6.1.22 Problem 22

Francisco rolls a fair die and records the value he rolls. Afterwards, he continues rolling the die until he obtains a value at least as large as the first roll. Let N be the number of rolls after the first he performs. Find $\mathbb{E}[N]$.

6.1.23 Problem 23

Fix positive integers n and k , and suppose $X_1, \dots, X_n \sim \text{Unif}(0, 1)$ IID. We say that there is an *increasing k -chain* starting from position i if $X_i < X_{i+1} < \dots < X_{i+(k-1)}$. Find n such that the expected number of increasing k -chains among $X_1, \dots, X_n$ is 1.

6.1.24 Problem 24

A genie rolls a fair 6-sided die. You can bet on the outcome of the die being even or odd. For every \$4 you bet on the die being even, you will receive \$6 back if correct. Similarly, for every \$6 you bet on the die being odd, you will receive \$9 back. Playing optimally, how much should you bet on this game?

6.1.25 Problem 25

Mike rolls three fair 6-sided dice at once. What is the expected median value of the dice?

6.1.26 Problem 26

Let X_1 and X_2 be two independent random variables with variance 25 and 100, respectively. Find the value of $0 \leq c \leq 1$ that minimizes the variance of $Y = cX_1 + (1 - c)X_2$.

6.1.27 Problem 27

Suppose you purchase a 1 share of stock A . A share of stock A has mean return 2 and return variation 4. You also want to purchase stock B . 1 share of B has mean return 3 and return variation 9. Additionally, the returns of A and B are correlated with coefficient $-1/2$. How many shares of stock B should you purchase so that the variance of your portfolio consisting of both A and B is minimum?

6.1.28 Problem 28

Pretend you have a simple options chain on the expected value of the product of two dice rolls. What would the call option at 30 be priced at in this market?

6.1.29 Problem 29

Pretend you have a simple options chain on the expected value of the product of two dice rolls. What would the put option at 4 be priced at in this market?

6.1.30 Problem 30

Let $N(t)$ be a Poisson Process with rate 5. Compute $\text{Cov}(N(5), N(15))$.

6.1.31 Problem 31

On average, 7 customers arrive at the QuantEssential gift shop per hour following a Poisson process. Suppose it takes 10 minutes to serve each customer. Compute the sum of the mean and variance of total service time in hours for customers arriving in a given 1 hour period.

6.1.32 Problem 32

Gabe makes rugs. The number of imperfections follows a Poisson distribution with an average of 4 per square yard. It costs 10 to repair each imperfection. Find the sum of the mean and variance of the repair cost for a 10-square-yard rug.

6.1.33 Problem 33

There is a game where an integer is picked uniformly between 2 and 10 inclusive. You win $\$n$ if the integer is prime and lose $\$ \frac{n}{2}$ if its composite (where n is the integer picked). How much would you pay to play this game? If you don't wish to play, enter 0.

6.1.34 Problem 34

Let A be a random 2×2 matrix formed by having each element ij set by $A_{ij} \sim \text{Bernoulli}(0.5)$. Find $\mathbb{E}[\det(A - A^T)]$.

6.1.35 Problem 35

You are given 8 fair coins and flip all of them at once. Afterwards, you are allowed to reflip as many coins as you would like one time each. At the end, you are given \$1 per head that appears. Assuming optimal play, find the fair value of this game.

6.1.36 Problem 36

Danielle is playing a game. She flips a fair coin repeatedly until she gets 2 heads in a row, after which she finishes the game. Determine the expected number of coin flips.

6.1.37 Problem 37

An basket containing 2 red and 1 blue orb is in front of you. Every turn you take an orb out uniformly at random and replace it with a blue orb, regardless of the color selected. Given you can always replace with blue orbs, find the expected amount of draws needed until the urn only contains blue orbs.

6.1.38 Problem 38

You are currently bidding for a painting. You know that the value of the painting is between \$0 and \$100,000 uniformly. If your bid is greater than the value of the painting, you win and sell it to an art museum at a price of 1.5 times the value. What's your bid to maximize your profit? If you can not profit, bid \$0.

6.1.39 Problem 39

On average, how many rolls of a fair 6-sided die must be rolled on average to get two 6s in a row?

6.1.40 Problem 40

Given that $\text{Var}(X) = 25$ and $\mathbb{E}[X] = 3$, find $\mathbb{E}[X^2]$.

6.1.41 Problem 41

You are sailing from one island to another. Uninterrupted travel between the two islands takes 12 minutes. However, there are 5 wave checks between the islands. Four of the wave checks will stop you with 25% probability each. These wave checks will add 1 minute to your total travel time if you are stopped. However, the last wave check stops you with 75% probability. This wave check will add 4 minutes to your travel time if you are stopped. Find the expected duration (in minutes) of the trip between the two islands.

6.1.42 Problem 42

Ab rolls a fair standard 6-sided die. Afterwards, given he rolls k on the first roll, Ab rolls a fair $(6 + k)$ -sided die for the second roll with the values $1, 2, \dots, 6 + k$. Find the expected value that appears on Ab's second roll.

6.1.43 Problem 43

Let $X_1, X_2, \dots, X_{40}$ be IID random variables with $\mathbb{E}[X_1] = 2$ and $\mathbb{E}\left[\frac{1}{X_1}\right] = 1$. Define $S_n = X_1 + \dots + X_n$. Compute $\mathbb{E}\left[\frac{S_{20}}{S_{40}}\right]$.

6.1.44 Problem 44

Juliana is playing "Fifteen" by Taylor Swift. The length of the song is 234 seconds. Juliana offers you the following bet: You may guess a value $0 < x < 1$. Juliana will then stop the song at a uniformly random point along its duration. If the point at which she stops the song is at most $234x$ seconds into the song, you must pay Juliana $2x$ monetary units. Otherwise, she pays you x monetary units. Find the value of x that maximizes your expected profit.

6.1.45 Problem 45

Adam is rolling a fair 10-sided die. He gets to roll repeatedly and may decide when to stop at any time. If he obtains a value that is not 10 on each roll, he adds the upface to his total monetary sum. If he rolls a 10, he loses all of his money. If Adam plays optimally, he should stop once he has at least \$ k total. Find k .

6.1.46 Problem 46

We are playing a game with two players. The first player rolls a 6-sided die 10000 times and the second player 10001 times. What are the odds of the second player getting more prime numbers than the first one?

6.1.47 Problem 47

Let $U \sim \text{Unif}(0, 1)$ and $X_1, X_2, \dots \sim \text{Unif}(0, 1)$ IID. Define $N = \min\{n \in \mathbb{N} : X_n > U\}$. Find $\mathbb{E}[N]$. Enter -1 if the answer is infinite.

6.1.48 Problem 48

Let $X, Y \sim \text{Unif}(3, 4)$ IID. Find $\mathbb{E}[|X - Y|]$.

6.1.49 Problem 49

Suppose that each card in a standard deck is given a value equal to it's face, with $A = 1, J = 11, Q = 12$, and $K = 13$. However, any card that is heart is doubled in value (ex: Ace of Hearts is worth 2). Find the expected value of a uniformly at random selected card from the deck.

6.1.50 Problem 50

On average, how many times do you need to roll a standard fair 6-sided die to observe all of the sides?

6.1.51 Problem 51

Suppose 20 people whose heights follow some unknown continuous distribution are arranged in a single-file line. We then stand at the front of the line and observe that we can see someone's head if they are taller than everyone that comes before. Let X be the number of visible heads. Compute $\mathbb{E}[X]$? Round to the nearest tenths.

6.1.52 Problem 52

You have made X letters for recruiters at X firms. However, you forgot which recruiter works for which firm, so you randomly assign each letter to a firm (only one letter going to each firm). What is the expected number of letters going to their correct firm?

6.2. Level 2

6.2.1 Problem 1

N employees are driving their individual cars on their way to work at QuantEssential. All cars are reasonably spaced apart, and they all travel at distinct speeds which are randomly assigned. When a faster car catches up to a slower car, it assumes the slower car's speed. After a very long period of time has passed, all N cars have settled into K clusters traveling at distinct speeds. What is the expected value of K when $N = 10$?

6.2.2 Problem 2

We select 3 points uniformly and randomly from the circumference of the unit circle. These 3 points divide the circle into 3 arcs. Find the expected length of the arc that contains the point $(1, 0)$. The answer is in the form $q\pi$ for a rational number q . Find q .

6.2.3 Problem 3

A random angle $\theta_1 \sim \text{Unif}(0, 2\pi)$ is selected. The arc of the unit circle that sweeps out θ_1 radians is marked red going counterclockwise starting from $(1, 0)$. Two other angles $\theta_2, \alpha \sim \text{Unif}(0, 2\pi)$ IID (independently, identically distributed) are also selected. Afterwards, an arc of length θ_2 radians starting from the point that is α radians counterclockwise of $(1, 0)$ is swept out and colored blue. What is the probability that the blue and red regions are disjoint.

6.2.4 Problem 4

Clarence is getting this bread. Literally. Clarence has 7 small loaves and 3 large loaves of bread in a bag. He draws in his bag and pulls out a loaf of bread uniformly at random, one-by-one. If it is a small loaf, Clarence eats it. If it is a large loaf, Clarence cuts it into two small loaves. He eats one one of them and puts the other one back in the bag. Find the expected number of small loaves that Clarence eats until the bag contains only small loaves left, inclusive of the small loaf he eats as a result of the last large loaf he draws.

6.2.5 Problem 5

On average, how many rolls of a fair 6-sided die are required to get two consecutive rolls with a difference of at most 1?

6.2.6 Problem 6

Every box of cereal contains one toy from a group of 5 distinct toys, each of which is mutually independent from the others and is equally likely to be within a given box. How many distinct toys can you expect to collect if you bought 7 boxes?

6.2.7 Problem 7

Consider a line of 20 adjacent colorless squares. Color in each individual square black or white with equal probability, independent of all other squares. A connected 5-component is a group of 5 consecutive black squares. Note that overlapping components are not counted. For example, WBBBBBWWB and WBBBBBBWB have 1 connected 5-component, but

WBBBBBBBBBBBW has 2 connected 5-components. Find the expected number of connected 5-components in our line. The answer can be written as a simplified fraction of the form $\frac{p}{q}$. Find $p + q$.

6.2.8 Problem 8

You uniformly select two random points from the circumference of the unit circle. Find the expected length of the chord (line segment) between the two points, with the answer is in the form $\frac{x}{\pi}$ for some rational number x . Find x .

6.2.9 Problem 9

Let C_n count the number of cycles in a random permutation of $\{1, 2, \dots, 2n\}$ that are larger than n in length. Compute $\lim_{n \rightarrow \infty} \mathbb{E}[C_n]$. The answer is in the form $\ln(q)$ for some rational number q . Find q .

6.2.10 Problem 10

14 pieces of paper labelled $1 - 14$ are placed in a line at random. Spot i is a local maximum if the paper at the i th position is strictly larger than all of its adjacent papers. Find the expected number of local maxima in the sequence. For example, with 6 numbers, 513246 has 3 local maxima at the first, third, and last spots.

6.2.11 Problem 11

You repeatedly draw marbles from a bag containing 50 red and 50 blue marbles until there are no more marbles left, recording the order of red and blue marbles drawn. You then count the number of "runs," where a run is defined as any number of consecutive marbles of the same color. For example, *RBRRRRBRR* contains 5 runs. What is the expected number of runs that you observe?

6.2.12 Problem 12

Say that you are flipping a fair coin where you can stop flipping whenever you want. Your goal is to maximize the ratio between the number of heads you get with the total number of flips. The ratios are in the form $a : 1$. What is the expectation of a when you follow the strategy of stopping when you have more heads than tails? Round your answer to the nearest thousandths.

6.2.13 Problem 13

You have 2 boxes and 100 distinct cards numbered 1 – 100. At each turn, you deal a card from the top of the deck and place the card in a box uniformly at random. What is the expected value of the smallest numbered card in the box that has card 100 in it? The answer is in the form $a(1 - a^{-b})$ for integers a and b . Find ab .

6.2.14 Problem 14

Two players, say A and B , play the following game: Both players have 100 marbles and may put anywhere between 1 and 100 marbles in the box each. This decision is not revealed to the other player. Then, they draw 2 marbles with replacement between trials. If the marble belongs to A , then assuming that A put a marbles in the box, A is paid $100 - a$ monetary units from a third party. Similarly if the marble belongs to B , then assuming B put b marbles in the box, B is paid $100 - b$ monetary units from a third party. Assume both players play optimally. Find the expected total payout of player A .

6.2.15 Problem 15

Two players, called X and Y , engage in a strategic game: Each has 100 tokens and secretly decides how many tokens, between 1 and 100, to contribute to a shared pool. Once both players have made their decisions, one token is randomly drawn from the pool. If the token belongs to X , then assuming X contributed x tokens, X receives $100 - x$ monetary units from Y . Similarly, if the token belongs to Y , then assuming Y contributed y tokens, Y receives $100 - y$ monetary units from X . Both players aim to maximize their respective monetary outcomes and are assumed to play optimally. How many tokens should X choose to contribute?

6.2.16 Problem 16

You are playing a one-player game with two opaque boxes. At each turn, you can choose to either "place" or "take". "Place" places \$1 from a third party into one box randomly. "Take" empties out one box randomly and that money is yours. This game consists of 100 turns where you must either place or take. Assuming optimal play, what is the expected payoff of this game? Note that you do not know how much money you have taken until the end of the game.

6.2.17 Problem 17

There are 100 noodles in your bowl of ramen. You take the ends of two noodles uniformly at random and connect the two, putting the connected noodle back into the bowl and continuing until there are no ends left to connect. On average, how many circles will you create? Round to the nearest whole number.

6.2.18 Problem 18

You generate a uniformly random number in the interval $(0, 1)$. You can generate additional random numbers as many times as you want for a fee of \$0.02 per generation. This decision can be made with the information of all of the previous values that have been generated. Your payout is the maximum of all the numbers you generate. Under the optimal strategy, find your expected payout of this game.

6.2.19 Problem 19

You select a uniformly random starting point on the circumference of a circle, say P_1 . Then, you select another uniformly random point on the circumference, P_2 , and draw a line segment between P_1 and P_2 . You then continue to select uniformly random points on the circumference of the circle, say $P_3, \dots, P_n$, and draw a line segment between P_i and P_{i+1} for each $1 \leq i \leq n-1$. Lastly, you draw a line segment from P_n back to P_1 . Find the expected number of intersections between line segments on the interior of the circle as a function of n . Evaluate this for $n = 12$.

6.2.20 Problem 20

7 orbs are labeled 1 – 7 and are linked linearly in a vertical stack from the ceiling with orb 1 being a part of the ceiling and orb 7 being closest to the floor. Each orb is attached to adjacent orbs by a chain link. At each time step, one of the remaining links is going to be uniformly at random selected and cut. As a result, all the orbs below that link will fall and shatter. What is the expected number of cuts needed until orb 1 is the only remaining orb?

6.2.21 Problem 21

Let $X_1, X_2, \dots$ be IID $\text{Unif}(0, 1)$ random variables and let $N = \min\{n : X_1 + \dots + X_n > \ln(2)\}$. Find $\mathbb{E}[N]$.

6.2.22 Problem 22

Let $X_1, X_2, \dots \sim \text{Unif}(0, 1)$ IID and $N_2 = \min\{n : X_1 + \dots + X_n > 2\}$. Find $\mathbb{E}[N_2]$. Your answer will be in the form $ae^2 + be$ for integers a and b . e here is Euler's constant. Find $a + b$.

6.2.23 Problem 23

Define $X_1, X_2, \dots \sim \text{Unif}(0, 1)$ IID $N_1 = \min\{n : X_1 + \dots + X_n > 1\}$, and $S_n = X_1 + \dots + X_n$. Compute $\mathbb{E}[S_{N_1}]$. The answer will be in the form ce for a rational number c . Find c .

6.2.24 Problem 24

A fair 5-sided die with the values 1 – 5 on the sides is rolled repeatedly until the sum of all upfaces is at least 5. Find the expected number of times the die needs to be rolled.

6.2.25 Problem 25

You are playing a game with a total of 100 actions and a fair 20-sided die. The die starts with face of 1. The two actions you can perform in the game are to roll and to take. Performing a roll re-rolls the current face of the die. Performing a take allows you to cash out the current face of the die. The game does ends when you complete all 100 actions and you must roll the die again before doing another take. Your strategy is to accept any number that is at least some threshold n . This n must be decided in advance and is fixed for the entire game. Assuming rational play in selecting n , find your expected payout.

6.2.26 Problem 26

You have 3 cards, each labeled n , $n + 1$, $n + 2$, and you don't know n . All cards start face down. You flip one card and observe the value. If you say "stay", your payout is equal to that card's value. If you don't "stay", then you flip another card. Again, choose to "stay" (and receive payout equal to the 2nd card's value) or flip the final card and receive payout equal to the final card's value. Design the optimal strategy. The expected out payout of this strategy is $n + c$ for some constant c . Find c .

6.2.27 Problem 27

Anna and Brenda are playing a game. They repeatedly toss a coin. Anna wins if 3 heads appear in a row. Brenda wins if 3 tails appear in a row. What is the expected number of coin tosses for a winner to be determined?

6.2.28 Problem 28

An 8-digit number is going to be created by a two-step process. First, select a uniformly random integer $k \in \{0, 1, \dots, 8\}$. Afterwards, we select one 8-digit number uniformly at random from the collection of 8-digit numbers with exactly k 1s and the other $8 - k$ digits are 9s. Call this selected number X . Find $\mathbb{E}[X]$.

6.2.29 Problem 29

We have $X \sim N(0, 1)$ and $Y \sim N(0, 4)$ are independent. Compute $\mathbb{E}[|Y - X|]$. The answer will be in the form $\left(\frac{K}{\pi}\right)^b$ for rational b and K . Find bK .

6.2.30 Problem 30

A bakery manager uniformly at random selects an integer (name it k) between 1 and 4, inclusive. He then chooses from k distinct desert types at his shop to create a gift basket consisting of k items. Find the expected number of ways that the manager can create the gift basket.

6.2.31 Problem 31

Alice rolls a fair 6-sided die with the values 1 – 6 on the sides. She sees that value showing up and then is allowed to decide whether or not she wants to roll again. Each re-roll costs \$1. Whenever she decides to stop, Alice receives a payout equal to the upface of the last die she rolled. Note that there is no limit on how many times Alice can re-roll. Assuming optimal play by Alice, what is her expected payout on this game?

6.2.32 Problem 32

Lebron is shooting free throws. He hits the first shot but misses the second. For $n \geq 3$, the probability of making the n th free throw is equal to the proportion of free throws he made during his first $n - 1$ attempts. If Lebron makes 100 attempts, how many freethrows can we expect him to make.

6.2.33 Problem 33

A $10 \times 10 \times 10$ cube is painted red on the surface and then cut into 1000 $1 \times 1 \times 1$ cubes and one is selected uniformly at random. Find the expected number of red faces on this cube.

6.2.34 Problem 34

Alice and Bob play the following game: Alice generates a uniformly random integer between 1 and 100, inclusive of both. Bob can pick any number that he wants between 1 and 100, inclusive of both. Whoever has the larger number must pay out the amount of the smaller number to the person with the smaller number. For example, if Alice selects 70 and Bob selects 50, Bob will receive 50 from Alice. If they select the same number, nobody pays out anything. Assuming optimal play by Bob, what is his expected payout?

6.2.35 Problem 35

Imagine you are waiting on a bus. The bus schedule is fixed, appearing every 10 minutes. Although, the driver independently between appearances, may want to refill on gas. The probability the driver refills on gas is 10% probability per trial, independently between trials. Note, if he fills up on gas, 1 hour is added to his travel time. If you arrive at a uniformly random time throughout the day, what is the expected time until the next bus appears, in minutes?

6.2.36 Problem 36

Suppose you have 20 cards with the values 1 – 10 each appearing twice in a deck. You draw 2 cards at a time uniformly at random from the 20 cards. If they match in value, you remove them from the deck. Otherwise, they are put back into the deck. The game finishes once there are no more cards to draw. Each drawing of two cards is a turn. Find the expected number of turns needed to finish the game.

6.2.37 Problem 37

n married couples (n husbands and n wives) sit at a round table of $2n$ seats at random. The seating scheme is such that the seats must alternate between men and women. Find the number of couples that will sit together $\mathbb{E}[N]$ on average. Assume $n > 2$.

6.2.38 Problem 38

On average, how many times does a fair 6-sided die need to be rolled to obtain two consecutive rolls that differ by exactly 1?

6.2.39 Problem 39

Every box of cereal contains one toy from a group of five distinct toys, each of which is mutually independent from the others and is equally likely to be within a given box. On average, how many boxes of cereal will you need to open in order to collect all five toys?

6.2.40 Problem 40

An electrician sets up 12 light bulbs on a circle. Each pair of light bulbs has a wire connecting them. Each individual wire is active with probability $\frac{1}{2}$, independent of all other wires. A collection of $2 \leq k \leq 12$ light bulbs is said to form a complete k -circuit if between any 2 light bulbs in the circuit, we have an active wire. A lightbulb can be part of multiple circuits. Find the expected number of 4-circuits in our electrician's arrangement of light bulbs.

6.2.41 Problem 41

You and your friend are playing a game with a fair coin, tossing it and writing down the outcomes. You win if HTH appears before HHT, else your friend wins. What is the probability that your friend wins?

6.2.42 Problem 42

Suppose that Z is a uniformly random point selected from the boundary of the unit circle in the complex plane $|z| = 1$. Compute $\mathbb{E}[Z^2]$.

6.2.43 Problem 43

An ant walks the corner of a 3D cube and moves to one of the three adjacent vertices with equal probability at each step. Find the expected number of steps needed for the ant to return to the vertex it started at.

6.2.44 Problem 44

After being caught in an embarrassing high-profile data fabrication scandal, Calambya University decides to stop sharing data with the national college ranking organization, effective in 2023. Rival school Mahogany University is overjoyed and offers each student a tuition refund worth X thousand dollars, where X = Calambya's 2023 Ranking – Mahogany's 2023 Ranking. One Mahogany student, Justin, offers to sell the following contract to his friend Kevin for 0.70 thousand dollars, before X is known: Justin will give Kevin the right but not the obligation to purchase Justin's tuition refund for only 0.5 thousand dollars. Suppose $X \sim \text{Exp}(\beta)$, where β is the scale. Compute β to the nearest tenth under the assumption that the contract is priced fairly.

6.2.45 Problem 45

Let $X_1, \dots, X_5$ be the upfaces of 5 independent rolls of a fair 6-sided die with values 1 – 6 on each side. Rank each of the following from largest to smallest in terms of value: a) $\mathbb{E}[X_1 X_2]$; b) $\mathbb{E}[X_1^2]$; c) $\mathbb{E}[X_{(3)}^2]$, where $X_{(3)}$ is the median of the 5 die rolls.

Let $a = 1, b = 2$, and $c = 3$. Answer with the integer corresponding to the concatenation of the order from largest to smallest. For example, if you believe $c > b > a$, answer with 321.

6.2.46 Problem 46

How many times do we have to roll a fair 6-sided die till we roll two numbers in a row that differ by 2?

6.2.47 Problem 47

Abd continually rolls a fair 6-sided die until he obtains his first 6. Compute the expected number of times Abd obtains the value 5 before he stops rolling.

6.2.48 Problem 48

Let $X_1, X_2, \dots$ be a sequence of IID random variables with some continuous PDF $f(x)$. Let N be the time at which the sequence stops decreasing i.e. the first value n such that $X_1 \geq X_2 \geq \dots \geq X_{n-1}$ but $X_{n-1} < X_n$. Find $\ln(\mathbb{E}[N])$.

6.2.49 Problem 49

On average, how many tosses of a fair coin does it take to see HTH ?

6.2.50 Problem 50

You're on a game show! The host tells you that \$200 is located behind 2 of 7 doors. The other 5 doors have nothing behind them. You have no idea which doors the money is behind. You keep selecting randomly until you find the first door with money behind it. You do not select any doors that you have selected prior. If each door opening costs \$ x and the game is fair, what is x ?

6.2.51 Problem 51

There are $2m$ members (where $m \geq 2$) seated randomly around a circular table at the Olympic Members Party. All $2m$ people have unique numbered labels $0, 1, \dots, 2x-1$, which represent how many people they know, the person with label i knows exactly i of the other $2x-1$ people at the table. These connections are one way, meaning that just because person B knows person A , it does not imply person A knows person B . What's the expected proportion of people that know at least one of the people on either side of them.

6.2.52 Problem 52

On average, how many cards in a normal deck of 52 playing cards do you need to flip over to observe your first ace?

6.2.53 Problem 53

4 fair coins are laid out in front of you and are flipped. You receive the amount of dollars equal to the number of heads that appear. However, you also have the option to re-flip all 4 coins at once one time for \$1. If you re-flip whenever you get 0 or 1 heads on the initial flip, what is the fair value of this game?

6.2.54 Problem 54

Suppose that you generate a uniformly random number in the interval $(0,1)$. You can generate additional random numbers as many times as you want for a fee of \$0.05 per generation. The number of additional generations must be made prior to your first number that you generate. Your payout is the maximum of all the numbers you generate. Assuming optimal strategy, what is your expected payout of this game?

6.2.55 Problem 55

We have an undirected graph with 11 nodes. For every node, you are able to access any other node (not including itself), all with an equal probability of $1/10$. Given this graph what is the expected number of steps to reach all nodes at least once (rounded to the nearest step)?

6.2.56 Problem 56

On average, how many flips of a fair coin will it take until you observe heads immediately followed by tails?

6.2.57 Problem 57

Imagine you are playing a game in which you draw four cards from a standard shuffled 52 card deck. What is the expected value of the highest card. Round your answer to the nearest hundredth. Note: Aces have a value of 1, Jacks a value of 11, Queens a value of 12, and Kings a Value of 13.

6.2.58 Problem 58

A factory wants to maximize the number of hours their employees are not working each year. If none of the employees has a birthday on a given day of the year, then every employee must come to work all 24 hours that day. If at least one person has a birthday on a given day, then none of the employees need to work that day. This factory has good work-life balance, so they want to hire the number of people that would maximize the expected total number of hours that employees are NOT working each year. How many employees should the factory hire? If the answer is infinite, answer -1 .

6.2.59 Problem 59

10 chords with uniformly randomly chosen endpoints are drawn on a circle. What is the expected number of intersections?

6.2.60 Problem 60

There is a box with 20 balls labeled from 1 through 20. You pick 10 balls out of the box. What is the expected maximum value of the 10 balls you picked out?

6.2.61 Problem 61

Let $X_1, X_2, X_3, X_4 \sim \text{Exp}(1)$ IID represent the lifetimes of 4 lightbulbs before burning out. Find the expected time until the last lightbulb burns out.

6.2.62 Problem 62

Suppose that $\ln(X) \sim N(0, 1)$. Find $\text{Var}(X^4)$. Your answer will be in the form of $e^a - e^b$ for positive integers a and b . Find $a + b$.

6.2.63 Problem 63

Suppose you have a rope that is 1 meter in length. You cut the rope uniformly at random along the length. Find the variance of the length of the longer piece (in meters).

6.2.64 Problem 64

6 distinct pairs of socks are in a drawer. You're in a hurry to pack for a trip, so you draw out 8 socks uniformly at random from the drawer. Find the expected number of pairs of socks that remain in the drawer.

6.2.65 Problem 65

Suppose you start with the 2×2 identity matrix I_2 . At each step, select one of the four elements of the matrix uniformly at random. If the element you select is a 1, change it to 0. If the element you select is a 0, change it to 1. Find the expected number of steps needed to obtain a singular matrix (i.e. the determinant is 0).

6.2.66 Problem 66

Let Y be any random variable only defined on $[-1, 1]$. Find the maximum possible value of $\text{Var}(Y)$.

6.2.67 Problem 67

Suppose that $A_1, \dots, A_{10}$ are independent random variables such that $\mu_i = \mathbb{E}[A_i] = 2i - 1$ and $\text{Var}(A_i) = 100$ meaning there is a common variance 100 but not a common mean.

Define $\bar{\mu} = \frac{1}{10} \sum_{i=1}^{10} \mu_i$. Compute $\mathbb{E}[(A_1 - \bar{A})^2]$.

6.2.68 Problem 68

Suppose that Steve in Minecraft is trapped in a dungeon and has a diamond pickaxe. There are 3 paths that he can take: Path 1 leads back to his original position after 2 days. Path 2 leads back to his original position after 4 days. Path 3 leads him to the surface after 8 days. Steve selects doors completely at random and independently between selections. Let N be the number of times Steve must select a door before reaching safety and T be the total amount of time it takes for Steve to get out. Furthermore, let X_i represent the amount of time that Steve takes to explore the path chosen at the i th turn. Let $A_n = \mathbb{E}[T \mid N = n]$ and $B_n = \mathbb{E}\left[\sum_{k=1}^n X_k\right]$. Find $\lim_{n \rightarrow \infty} \frac{A_n}{B_n}$. If the limit doesn't exist, enter -1 .

6.2.69 Problem 69

Consider selecting 20 independent and identically distributed (IID) points on the circumference of the unit circle, labeled $X_1, \dots, X_{20}$. We want to determine the expected arc length (in degrees) between X_1 and the closest point to it on the circle.

6.2.70 Problem 70

Carter has come into contact with a bounty of gold. He takes it to an auction shop. The auction shop says that each person that bids will place a bid that is uniformly distributed between \$500 and \$1000. They also state that they can recruit people to bid for a price of \$5 per person. Assuming Carter selects the optimal number of people to bid on his gold, what is his expected payout?

6.2.71 Problem 71

Consider the set $\Omega = \{1, 2, \dots, 20\}$. Two subsets A and B of Ω are uniformly at random selected from the power set of Ω independently. It is possible that $A = B$. Define $N = |A \cap B|$. Compute $\frac{\text{Var}(N)}{\mathbb{E}[N]}$.

6.2.72 Problem 72

At a party with 50 guests, each guest writes their name on a piece of paper and places it in a hat. Guests take turns drawing a name from the hat. Each guest forms a group with the person whose name they draw. If a guest draws their own name, they form a group alone. If a guest draws another guest's name and this continues in a cycle, they form a closed group. If Guest A pulls out Guest B 's name, Guest B pulls out Guest C 's name, and Guest C pulls out Guest A 's name, they are all a part of the same closed group and no one else will be able to join them. What is the average group size? Round your answer to the nearest tenth.

6.2.73 Problem 73

You have been arrested and need to post bail. You have a bag of \$100 bills of which one half are real and the other half are fake. You need to pay \$500 for bail. Every time you give the jail a fake bill, the funds you have given are given back to you and you need to start again. How many bills will you have to pull out of the bag on average? Assume the probability of drawing each type of bill is constant per trial.

6.2.74 Problem 74

You have 2 indistinguishable urns in front of you. The first urn has 4 (\$1 chips) and 6 (\$10 chips), while the second urn has 3 (\$1 chips) and 7 (\$10 chips). You reach into one urn at random and select a (\$1 chip). Then you have the opportunity to pick another chip at random either from same urn you picked the first chip from or at random from the other urn. Your payout is the value of the second chip you select. Under optimal gameplay, what is your expected payout?

6.2.75 Problem 75

A dealer at a casino holds a game that is played as follows: The player first pays \$ d to the dealer, where d is some positive integer less than 100. The dealer then intends to flip a coin d times. If at any point two consecutive heads are flipped, then the dealer stops flipping the coin, and the player wins and is awarded \$100. Otherwise, the player loses (and is awarded nothing). A player pays the dealer \$29. The dealer weights the coin before flipping it, changing the probability of flipping heads such that the player's expected net profit is non-positive. The player is not aware of this re-weighting. Find the maximum probability that the dealer should choose.

6.2.76 Problem 76

Suppose that M and S are IID $\text{Exp}(1)$ random variables. Given that $M = \mu$ and $S = \sigma$, $(X, Y) \sim \text{BVN}(\mu, \mu, \sigma^2, \sigma^2, \rho)$, where $|\rho| \leq 1$ is fixed. Compute $\text{Var}(Y)$.

6.2.77 Problem 77

Audrey repeatedly draws cards from a standard 52-card deck with replacement. Compute the expected number of draws for Audrey to get at least one card from each suit.

6.2.78 Problem 78

Siblings Alice and Bob play a game with marbles. Each player has one red and one blue marble and shows one marble to the other uniformly at random. If both show blue, Alice wins \$1. If both show red, Alice wins \$3. Else, Bob wins \$2. Note that the winnings come from their mother, not the other player. Let A_r and B_r define the ratio between the expected return and variance of Alice's and Bob's payoffs, respectively. What is $B_r - A_r$?

6.2.79 Problem 79

Martin is riding his remote control skateboard around campus. Assume his house is the origin $(0, 0)$, and that movement to the right is positive horizontally, and movement into campus (upwards) is positive vertically. Each second, Martin uniformly picks a random angle between 0 and 2π , moving the skateboard 1 foot in that direction. After 16 seconds, what is the expected squared distance from the origin?

6.2.80 Problem 80

We have a straddle strategy with strike $K = 0$. The underlying asset price is $S \sim N(0, 1)$. First, find the expected value of the straddle, v , then given that v^2 can be written in the form $\frac{a}{\pi}$ for a rational number a . Find a .

6.2.81 Problem 81

You are given a fair 20-sided die and 100 actions in a game. The die starts with upface 1. The two options you can perform are to roll and to take. Performing a roll re-rolls the current upface of the die. Performing a take allows you to cash out the current upface of the die. Note that the game does not end when you perform a take and that you do not have to roll between takes. Therefore, for example, you can just perform 100 takes on the initial \$1 upface and walk away with \$100 guaranteed. Your strategy is to cash out the upface when you roll at least some threshold n for the first time. You fix this n at the beginning of the game. Assuming rational strategy in selecting n , what is your expected payout on this game?

6.2.82 Problem 82

You are given a fair 6-sided die and you roll it. You can either choose to keep your roll and receive the observed value in dollars. Alternatively, you are allowed to roll again, but if the sum of your two rolls is at least 7, you pay the value equal to your first roll. If the sum of the two rolls is less than 7, you receive the second observed value in dollars. Assuming optimal play, what is your expected payout?

7. Limit Theorems

7.1. Level 1

7.1.1 Problem 1

Jon loves cheese. He decides to make 100 blocks of cheese. The distribution of the weight (in grams) of each block he makes follows IID $\text{Exp}\left(\frac{1}{250}\right)$ distribution. Let W_i denote the weight of the i th block of cheese, and T_{100} represent the total weight of the 100 blocks of cheese. Using Chebyshev's Inequality, what is an upper bound on $\mathbb{P}[T_{100} > 30000]$?

7.2. Level 2

7.2.1 Problem 1

Let X be a random variable with MGF $M(\theta)$ such that $M(1) = 7$. Find the smallest possible value for $M(4)$.

Acknowledgment The problems are not created by me. They are collected from online free resources, and organized into sections, and levels that I think are appropriate for my students. If you find any errors in the questions, please let me know, I will correct them.